

# Positivity of Serre’s Intersection Multiplicity

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September 23, 2026

## Abstract

We prove Serre’s positivity conjecture, including ramified mixed characteristic. If two nonzero finitely generated modules over a regular local ring have tensor product of finite length and complementary dimensions, then their intersection multiplicity is strictly positive.

## 1 Introduction

Let  $(R, \mathfrak{m})$  be a commutative Noetherian regular local ring of dimension  $d$ . For finite  $R$ -modules  $M, N$  whose tensor product has finite length, Serre’s intersection multiplicity is the integer

$$\chi^R(M, N) = \sum_{i=0}^d (-1)^i \text{length}_R \text{Tor}_i^R(M, N).$$

The support condition makes every term finite, and regularity makes the higher Tor groups vanish. This Euler characteristic measures the local intersection of the supports while accounting for higher Tor. Its positivity is not a formal consequence of the nonzero degree-zero term: the higher terms enter with alternating signs.

**Theorem 1.1.** *Let  $(R, \mathfrak{m})$  be a commutative Noetherian regular local ring of dimension  $d$ , and let  $M, N$  be nonzero finitely generated  $R$ -modules. If  $\text{length}_R(M \otimes_R N) < \infty$  and  $\dim_R M + \dim_R N = d$ , then*

$$\chi^R(M, N) > 0.$$

Here regularity means that  $\mathfrak{m}$  can be generated by  $d$  elements, and module dimension means the Krull dimension of its support. Theorem 1.1 proves Serre’s positivity conjecture without a restriction on characteristic or ramification. In particular, it does not require either module to lift to an unramified regular local ring.<sup>1</sup>

### 1.1 Background and significance

Serre introduced intersection multiplicities through the Euler characteristic of Tor. His reduction to the diagonal proves positivity in equicharacteristic and in the unramified setting; the same argument also treats formal power series rings over an arbitrary complete discrete valuation ring [25, Chapter V]. Thus ramification alone does not separate the known cases from the general problem. Roberts and, independently, Gillet–Soulé proved vanishing when the tensor product has finite length

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<sup>1</sup>A public post by Editan, dated December 9, 2025 and credited to Editan, Copilot, and Gemini, also asserts the prime-quotient form of Serre’s positivity conjecture. We record this as a same-scope claim, not as an established theorem; see [the author’s post](#).

and the sum of dimensions is smaller than the ring dimension [22, 23, 11]. Their arguments use local Chern characters and Adams operations, respectively. Gabber proved nonnegativity in the remaining mixed-characteristic case using alterations and an intersection calculation on a normal bundle [2, Section 6.1, Theorem 6.1.1]. Vanishing removes lower-dimensional contributions in the reductions below. For complementary-dimensional pairs, strict positivity must additionally rule out zero.

Further progress in ramified mixed characteristic includes Skalit's positivity theorem for regular local rings essentially smooth over a two-dimensional regular local base [26]. KC and Soto Levins prove positivity for modules satisfying a Serre-liftability condition [15, Theorem A]. In their positivity statement,  $R = S/(f)$  for an unramified regular local ring  $S$  and  $f \in \mathfrak{m}_S \setminus \mathfrak{m}_S^2$ , and  $M$  has a Serre lift  $L$ : a finite  $S$ -module with  $M = L \otimes_S R$  and  $\dim S - \dim L = \dim R - \dim M$ ; it need not satisfy the usual Tor-vanishing condition for a lift. Theorem 1.1 imposes neither this liftability condition nor the smoothness hypothesis.

The proof combines normalized-length methods with algebraic  $K$ -theory. Faltings introduced the normalized-length formalism in almost mathematics [9, Appendix 2]; Gabber–Ramero developed its almost-ring-theoretic foundations further [10]. The perfectoid length results of Cai–Lee–Ma–Schwede–Tucker [6], together with their characteristic- $p$  counterparts [19], give algebras on which parameter Koszul complexes are exact in positive degrees after passing to normalized length. The mixed-characteristic construction uses perfectoidization and almost purity, as developed by Bhatt–Scholze [4]. Land–Tamme's descent theorem for truncating invariants [18] supplies the categorical descent input. Bhatt–Hochster–Ma show that the existence of  $\lim$  Cohen–Macaulay sequences for both quotient domains implies positivity [3, Theorem 6.1]. Ishiro–Shimomoto obtain such sequences of algebras from perfectoid towers over complete mixed-characteristic Noetherian local domains with  $F$ -finite residue field [13, Theorem 3.17]. These are conditional routes through finite modules. Here no such tower on either quotient domain is assumed: the proof uses bounded complexes modulo free sequences of negligible normalized rank.

## 1.2 Proof strategy

After standard reductions, it is enough to consider two complete local domains  $D = R/P$  and  $E = R/Q$  of positive dimensions  $h, l$ , where  $h + l = d$  and  $P + Q$  is  $\mathfrak{m}$ -primary. Each domain has a finite extension, denoted  $D'$  or  $E'$ , with a finite Galois action over a regular normalization base  $A_D$  or  $A_E$ , that is, a regular local subring over which the extension is finite. Associated algebras  $C_D, C_E$  carry additive nonnegative lengths  $\lambda_D, \lambda_E$ . Their parameter quotients have positive length, whereas positive parameter Koszul homology has length zero.

There are two distinct comparisons in the argument. First, rational  $K$ -theory with support at the closed point shows that ordinary and normalized Euler lengths agree after summing over all Galois conjugates. Write  $G_D, G_E$  for the two groups, and  $a_D = \text{rank}_D D'$ ,  $a_E = \text{rank}_E E'$ . For  $\sigma \in G_D$ , let  $c_\sigma$  be the normalized Tor Euler characteristic of  $C_D$  and  $E'$  over  $R$ , where the  $R$ -action on  $C_D$  is twisted by  $\sigma$ . The first comparison identifies the sum of the  $c_\sigma$  with  $|G_D| a_D a_E \chi^R(D, E)$ . It therefore suffices to show that each  $c_\sigma$  is positive. Fix one such twist.

To analyze that value, choose a regular sequence  $x$  in  $Q$  that is a parameter system on  $D$ . The module  $U = C_D/xC_D$  has positive finite normalized length over  $B = R/(x)$ . It can be approximated by finite-length  $B$ -modules  $M_n$  killed by one fixed power of the maximal ideal of  $B$ , with ordinary lengths divided by integers  $s_n$  converging to the relevant normalized lengths. Their minimal resolutions have ranks  $o(s_n)$  above degree  $l$ . Truncating at degree  $l$  therefore produces supported complexes over  $E'$  in the category that kills free sequences whose ranks divided by  $s_n$  tend to zero.

The second comparison takes place in this category. Its supported objects have nonnegative homology lengths, even when support appears only after passing to the quotient. The two Euler evaluations again agree on sums of conjugates. Evaluation using  $C_E$  leaves only degree zero, whose length is bounded below by a positive constant times the normalized number of generators of  $M_n$ . The common annihilator and the positive limiting length of  $M_n$  make this lower bound strictly positive. The second Galois average proves positivity for each value in the first average, completing the argument.

Figure 1 records the two interfaces. Here  $b_{\sigma,\gamma}$  denotes the  $C_E$ -evaluation of the truncated sequence after the additional twist  $\gamma \in G_E$ . Choose  $a$  with  $\mathfrak{m}_B^a M_n = 0$  for all  $n$ , and write  $\beta_0(M_n)$  for the minimal number of generators of  $M_n$ . Then  $\text{length}_B M_n \leq \beta_0(M_n) \text{length}_B(B/\mathfrak{m}_B^a)$ . Thus the positive normalized length of the  $M_n$  forces a positive normalized number of generators; it is this uniform estimate that makes every  $b_{\sigma,\gamma}$  strictly positive.

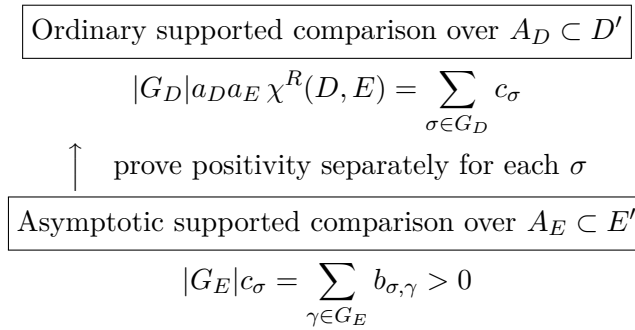


Figure 1: Two norm comparisons, not invariance of individual evaluations. The first is Lemma 5.2; the second is Equation (5.18). The uniform annihilator and generator bound give the strict inequality in Lemma 5.6.

The reusable technical points are the finite-surjection descent argument with arbitrary connective algebra coefficients and the construction of supported Euler evaluations after quotienting by negligible ranks. In particular, the comparison is proved on idempotent summands in the quotient, not only on complexes that already have uniform support before taking the quotient. No invariance of a normalized-length function under either Galois group is assumed.

Section 2 establishes the precise normalized-length inputs. Section 3 proves supported rational descent and the first Euler comparison. Section 4 constructs the rank quotient and its supported evaluations. Section 5 performs the approximation and the two comparisons to prove Theorem 1.1.

### 1.3 Consequences

Theorem 1.1 also has consequences for symbolic powers. For a ramified mixed-characteristic regular local ring  $(R, \mathfrak{m})$  and primes  $P, Q$  with  $\sqrt{P+Q} = \mathfrak{m}$  and  $\text{ht } P + \text{ht } Q = \dim R$ , the implication of Kurano–Roberts [16, Theorem 3.2] now gives

$$P \cap Q^{(r)} \subseteq \mathfrak{m}^{r+1} \quad (r \geq 1), \quad Q^{(r)} := Q^r R_Q \cap R.$$

Here ramified means that the residue characteristic  $p$  lies in  $\mathfrak{m}^2$ . This is a direct application of their positivity-to-symbolic-power theorem, rather than an additional ingredient in our proof.

The next consequence applies Skalit’s finite-support blowup argument [26, Corollary D] with Theorem 1.1 as its positivity input. Dutta studied the lower bound in this finite-support setting [8]. We include the argument to exhibit how strict positivity at the points of the blowup controls equality.

**Corollary 1.2** (Finite strict-transform intersection). *Let  $(A, \mathfrak{m})$  be a commutative Noetherian regular local ring with residue field  $k$ , and let  $P, Q$  be prime ideals such that  $A/(P + Q)$  has finite length and  $\dim(A/P) + \dim(A/Q) = \dim A$ . Write  $e_{\mathfrak{m}}$  for Hilbert–Samuel multiplicity and put*

$$T = \mathrm{gr}(A/P) \otimes_{\mathrm{gr} A} \mathrm{gr}(A/Q),$$

*where all associated graded rings use the maximal-ideal filtrations. If  $\dim T \leq 1$ , then*

$$\chi^A(A/P, A/Q) \geq e_{\mathfrak{m}}(A/P)e_{\mathfrak{m}}(A/Q),$$

*with equality if and only if  $\dim T = 0$ .*

*Proof.* If  $\dim A = 0$ , then  $A = k$ ,  $P = Q = 0$ , and both sides are 1 with  $T = k$ . Otherwise, let  $\tilde{X}$  be the blowup of  $X = \mathrm{Spec} A$  at its closed point, and let  $\tilde{Y}, \tilde{Z}$  be the strict transforms of  $Y = \mathrm{Spec}(A/P)$  and  $Z = \mathrm{Spec}(A/Q)$ . By [27, Lemma 3.5], the underlying set  $S$  of  $\tilde{Y} \cap \tilde{Z}$  identifies with that of  $\mathrm{Proj} T$ . Thus  $S$  is a finite set of closed points of the exceptional divisor, empty exactly when  $\dim T = 0$ . Fulton’s blowup formula and Skalit’s finite-support calculation [27, Theorem 3.4 and proof of Proposition 3.10] give

$$\chi^A(A/P, A/Q) - e_{\mathfrak{m}}(A/P)e_{\mathfrak{m}}(A/Q) = \sum_{x \in S} [\kappa(x) : k] \chi^{B_x}(M_x, N_x),$$

where  $B_x = \mathcal{O}_{\tilde{X}, x}$ ,  $M_x = \mathcal{O}_{\tilde{Y}, x}$ , and  $N_x = \mathcal{O}_{\tilde{Z}, x}$ . Each residue degree is finite and positive since the exceptional divisor is projective over  $k$ . The blowup  $\tilde{X}$  is regular [14, Proposition A.4(1)]. The modules  $M_x, N_x$  are nonzero and finite over  $B_x$ , with finite-length tensor product because their intersection is supported on  $S$ . The local domains  $A/P$  and  $A/Q$ , as well as  $A$ , are formally equidimensional and universally catenary. Thus [27, Lemma 3.8 and proof of Proposition 3.10] gives  $\dim M_x = \dim(A/P)$ ,  $\dim N_x = \dim(A/Q)$ , and  $\dim B_x = \dim A$ . Theorem 1.1 therefore makes every local intersection multiplicity in the sum strictly positive. The asserted inequality follows, and equality holds exactly when  $S$  is empty.  $\square$

## 1.4 Conventions

We fix universes large enough for the module and sequence categories under consideration. Complexes are indexed homologically. An associative derived algebra is *connective* when its homotopy groups in negative degrees vanish. Perfect modules over associative algebras are right modules. All scalar localizations of coefficient algebras are along the central image of the commutative base. We use nonconnective algebraic  $K$ -theory of perfect complexes; a subscript  $\mathbb{Q}$  denotes rationalization. For a sequence  $a = (a_1, \dots, a_r)$ , write  $K(a; V)$  for its homological Koszul complex with coefficients in  $V$ , and put  $H_i(a; V) = H_i(K(a; V))$ . The notation  $a^v$  denotes  $(a_1^v, \dots, a_r^v)$ , not the  $v$ th power of the ideal it generates. Normalized lengths are allowed to vanish on nonzero modules; all uses of their additivity involve finite values or nonnegative extended values for which the indicated operations are defined.

## 2 Residue fields and normalized length

The comparison arguments require algebras on which positive parameter Koszul homology has length zero and parameter quotients have prescribed positive length. The finite approximations in Section 5 also require that this length be defined on underlying modules over a flat tower, without an action of the full algebra. We construct these objects after arranging a perfect residue field;

neither unramifiedness nor Cohen–Macaulayness of the domain is required. Both characteristics are needed: a prime quotient of the mixed-characteristic ambient ring may itself have characteristic  $p$ , even though ambient equicharacteristic positivity is already known.

## 2.1 Extending the residue field

**Lemma 2.1** (Residue-field extension). *Let  $(B, \mathfrak{b}, k)$  be a complete Noetherian local ring, and let  $\bar{k}$  be an algebraic closure of  $k$ . There is a complete Noetherian local ring  $(B', \mathfrak{b}', \bar{k})$  and a flat local map  $B \rightarrow B'$  such that  $\mathfrak{b}B' = \mathfrak{b}'$ . For every finite  $B$ -module  $V$  and every  $a \geq 1$ ,*

$$\text{length}_{B'}((V \otimes_B B')/(\mathfrak{b}')^a(V \otimes_B B')) = \text{length}_B(V/\mathfrak{b}^a V).$$

*Consequently module dimensions are preserved, as are finite lengths. If  $B$  is regular, then  $B'$  is regular. For finite  $B$ -modules  $M, N$  whose tensor product has finite length, every Tor length, and hence their intersection multiplicity when defined, is preserved.*

*Proof.* In equal characteristic, a coefficient field gives a presentation  $B = k[[X_1, \dots, X_t]]/I$ . Put  $B' = \bar{k}[[X_1, \dots, X_t]]/I\bar{k}[[X_1, \dots, X_t]]$ . The map of power-series rings is flat, by the local flatness criterion for the variables and the flat coefficient-field extension. Its base change to the quotient has the asserted residue field and maximal ideal.

In the remaining case the residue characteristic is a prime  $p$ . Cohen structure supplies a presentation  $B = V[[X_1, \dots, X_t]]/I$ , where  $V$  is a complete discrete valuation ring with uniformizer  $p$  and residue field  $k$ . We construct a discrete valuation extension with unchanged uniformizer and residue field  $\bar{k}$ , without assuming that  $k$  is perfect. Given a simple algebraic residue-field extension, lift its monic irreducible polynomial to  $f \in V[X]$ . The algebra  $V[X]/(f)$  is finite free over  $V$ , its reduction modulo  $p$  is the desired field, and all its maximal ideals lie over  $(p)$ . It is therefore local with maximal ideal generated by  $p$ . Since  $p$  is a nonzerodivisor, it is a one-dimensional regular local ring, hence a discrete valuation ring. This argument also applies to inseparable residue extensions.

Perform this construction along a tower of simple algebraic extensions exhausting  $\bar{k}$ , taking unions at limit stages. In every such union, a nonzero element is a unit times a nonnegative integral power of  $p$ . Every nonzero ideal is generated by an element of least valuation, so the union is still a discrete valuation ring. Its completion  $V'$  has uniformizer  $p$  and residue field  $\bar{k}$ . The map  $V \rightarrow V'$  is flat because  $V'$  is torsion-free over  $V$ . Now take

$$B' = V'[[X_1, \dots, X_t]]/IV'[[X_1, \dots, X_t]].$$

The same local flatness criterion proves the required flatness. This is the usual residue-extension construction; compare [19, Lemma 2.2].

For either construction, tensor a composition series of a finite-length  $B$ -module with  $B'$ . Flatness preserves exactness, and every simple factor becomes  $B'/\mathfrak{b}'$ . This proves the length assertions and, applied to  $V/\mathfrak{b}^a V$ , the displayed identity. Equality of Hilbert–Samuel functions gives equality of module dimensions. If  $B$  is regular of dimension  $d$ , its  $d$  maximal-ideal generators generate  $\mathfrak{b}'$ , and  $\dim B' = d$ , so  $B'$  is regular. Finally flat base change identifies the Tor modules after extension; their finite lengths are preserved by the preceding argument.  $\square$

## 2.2 Regular bases and the tower length

For a perfect field  $k$  of characteristic  $p$ , let  $W(k)$  denote its ring of Witt vectors.

**Lemma 2.2** (Regular normalization bases). *Let  $(D, \mathfrak{d}, k)$  be a complete Noetherian local domain of dimension  $h > 0$ , where  $k$  is perfect of characteristic  $p > 0$ . There is a finite, generically separable inclusion  $A \subseteq D$ , inducing the identity on residue fields, with*

$$A = k[[t_1, \dots, t_h]] \quad \text{if } \text{char } D = p, \quad A = W(k)[[t_2, \dots, t_h]] \quad \text{if } \text{char } D = 0.$$

*In the second case set  $t_1 = p$ . In both cases  $t_1, \dots, t_h$  is a parameter system of  $D$  and a regular parameter system of  $A$ .*

*Proof.* The positive-characteristic assertion is the Cohen–Gabber normalization theorem [17]. In mixed characteristic, choose a coefficient ring  $W(k) \subseteq D$  and extend  $p$  to a parameter system  $p, t_2, \dots, t_h$ . Completeness and the parameter condition make the resulting power-series map module finite. Its kernel is zero: the source and target have dimension  $h$ , whereas a nonzero ideal of the regular local domain  $W(k)[[t_2, \dots, t_h]]$  lowers dimension. The fraction-field extension is separable in characteristic zero.  $\square$

Fix one of the bases in Lemma 2.2 and compatible  $p$ -power roots of its parameters. Write

$$A_n = A[t_1^{1/p^n}, \dots, t_h^{1/p^n}], \quad \Lambda = \varinjlim_n A_n. \quad (2.1)$$

The tower is uncompleted. Only in mixed characteristic do we also use its  $p$ -adic completion, denoted  $A_\infty$ .

**Lemma 2.3** (The tower and its length). *Each  $A_n$  is a complete regular local ring with residue field  $k$ . For  $b \geq a$ , the map  $A_a \rightarrow A_b$  is finite free of rank  $p^{(b-a)h}$ ; in particular  $A \rightarrow A_n$  has rank  $p^{nh}$ . On the category of  $\Lambda$ -modules killed by some power of  $\mathfrak{m}_A$  there is a nonnegative additive function  $\lambda_\infty$  with the following properties.*

1. *A finitely presented module descends to a finite-length  $A_n$ -module  $V_n$ , and*

$$\lambda_\infty(V_n \otimes_{A_n} \Lambda) = p^{-nh} \text{length}_{A_n}(V_n). \quad (2.2)$$

2. *For a finitely generated module  $V$ , its length is the infimum of the lengths of finitely presented modules in this category admitting a surjection onto  $V$ .*
3. *For an arbitrary module  $V$  in this category,*

$$\lambda_\infty(V) = \sup\{\lambda_\infty(V_0) : V_0 \subseteq V \text{ is finitely generated}\}. \quad (2.3)$$

*The value on an arbitrary module may be infinite. In mixed characteristic, restriction from  $A_\infty$  to  $\Lambda$  gives the same length on modules killed by a fixed power of  $\mathfrak{m}_A$ .*

*Proof.* In characteristic  $p$ , the stages are power-series rings in the chosen roots. In mixed characteristic they are power-series rings over the complete discrete valuation ring  $W(k)[p^{1/p^n}]$ . The monomials in the successive roots, with each exponent less than the relevant root degree, form bases for all transition extensions. This proves the assertions about regularity, ranks, and residue fields.

Finite presentations over the union descend to a stage. The additional relations expressing annihilation by a fixed power of  $\mathfrak{m}_A$  also descend after increasing that stage, because there are finitely many generators and relations to consider. The resulting finite-stage module has finite length. Formula (2.2) is unchanged on enlarging the stage, by flatness and the computed free ranks. The infimum and supremum extensions, including their additivity and compatibility with this formula,

are Faltings' normalized-length formalism [9, Appendix 2], in the formulation of [6, Definition 2.2.1 and Propositions 2.2.3, 2.2.6(a)]. The same residue field implies that  $A_n$ -length and  $A$ -length agree on finite-length  $A_n$ -modules.

In mixed characteristic  $\Lambda$  is  $p$ -torsion-free, and the natural map induces  $\Lambda/p^a\Lambda \simeq A_\infty/p^aA_\infty$  for every  $a \geq 1$ . A module killed by  $\mathfrak{m}_A^a$  is killed by  $p^a$ . Its submodules and finite generating sets are therefore identical over the two bases. In computing the infimum over finitely presented covers of such a module, one may replace a cover  $V$  by  $V/\mathfrak{m}_A^aV$ ; this is still finitely presented. These covers also correspond over the identical quotient rings, proving the last assertion.  $\square$

**Lemma 2.4** (Zero-length limits). *Suppose all modules under consideration are killed by one fixed power of  $\mathfrak{m}_A$ . Sums, direct sums, and directed colimits of zero-length modules have length zero. For an increasing union,*

$$\lambda_\infty\left(\bigcup_a V_a\right) = \sup_a \lambda_\infty(V_a).$$

*If  $(V_a)_{a \geq 1}$  is a direct system with  $\lambda_\infty(V_a) \rightarrow 0$ , then  $\lambda_\infty(\varinjlim_a V_a) = 0$ , without any injectivity assumption on the transition maps.*

*Proof.* A finite set of elements of a direct sum is contained in a finite subsum. Additivity and (2.3) give the assertion for direct sums; sums and directed colimits are quotients of direct sums. For an increasing union, every finite generating set lies in one stage. Finally a finitely generated submodule of  $\varinjlim_a V_a$  is contained in the image of every sufficiently late  $V_a$ , by lifting its generators to a common stage. Monotonicity bounds its length by  $\lambda_\infty(V_a)$  at all such stages. Apply (2.3). The common annihilator ensures throughout that these modules belong to the length category.  $\square$

### 2.3 Length algebras and parameter homology

The tower length will now measure modules over an algebra of the given domain. Its parameter identities have two uses: the base parameters control the Euler comparisons, while arbitrary parameter systems give the positive quotients used in Section 5. For an ideal  $I$  primary to the maximal ideal of a local ring  $D$ , write  $e(I, D)$  for its Hilbert–Samuel multiplicity.

Related normalized-length constructions, in which parameter-quotient lengths recover multiplicities and positive parameter Koszul homology has normalized length zero, appear in [21, Theorem 3.1]. Here we retain the chosen base  $A \subseteq D$  and the finite-stage, infimum, and supremum properties on underlying tower modules.

**Theorem 2.5** (Normalized-length input). *Let  $(D, \mathfrak{d}, k)$  and a chosen finite generically separable base  $A \subseteq D$  be as in Lemma 2.2, and let  $\Lambda$  be the tower (2.1). There is a  $D$ -algebra  $C_D$ , carrying a compatible  $\Lambda$ -action, such that restriction of scalars defines an additive nonnegative length  $\lambda_D = \lambda_\infty$  on its modules annihilated by a power of  $\mathfrak{d}$ . For every parameter system  $z_1, \dots, z_h$  of  $D$  and every integer  $v \geq 1$ ,*

$$\lambda_D H_i(z_1^v, \dots, z_h^v; C_D) = 0 \quad (i > 0), \tag{2.4}$$

$$\lambda_D(C_D/(z_1^v, \dots, z_h^v)C_D) = v^h e((z_1, \dots, z_h), D). \tag{2.5}$$

*The right side is positive and finite. The algebra and length are independent of the parameter system  $z$ . For the regular parameters of  $A$ , in particular,*

$$\lambda_D(C_D/(t_1^v, \dots, t_h^v)C_D) = v^h \text{rank}_A D. \tag{2.6}$$

*The finite-stage, infimum, and supremum properties of Lemma 2.3 remain available on underlying tower modules; they do not require that the relevant submodules or covers be  $C_D$ -modules.*

*Proof.* We give the positive-characteristic construction and then specify the mixed-characteristic literature inputs.

Assume first that  $D$  has characteristic  $p$ , and put  $C_D = D_{\text{perf}} = \bigcup_n D^{1/p^n}$  inside a fixed algebraic closure of its fraction field. Here  $\Lambda = A_{\text{perf}}$ . Write  $K = \text{Frac}(A)$ ,  $L = \text{Frac}(D)$ , and  $K' = K_{\text{perf}}$ . For  $q = p^n$  put

$$B_q = \Lambda \otimes_{A^{1/q}} D^{1/q}.$$

Flatness of  $\Lambda$  over  $A^{1/q}$  and torsion-freeness of  $D^{1/q}$  embed this tensor product into  $K' \otimes_{K^{1/q}} L^{1/q}$ . Generic separability makes the latter a field, by linear disjointness from a purely inseparable extension. Thus  $B_q = \Lambda D^{1/q} \subseteq C_D$ , and these subrings increase to  $C_D$ .

Let  $H$  be any  $\mathfrak{d}$ -primary ideal and set  $V_q = B_q/HB_q$  and  $W_q = \text{im}(V_q \rightarrow C_D/HC_D)$ . Write  $H^{[q]} = (a^q : a \in H)$  for its Frobenius power. Finite-stage normalization and the  $q$ -th-power isomorphism give

$$\lambda_\infty(V_q) = q^{-h} \text{length}_D(D/H^{[q]}).$$

Although  $B_q \subseteq C_D$ , the map on these quotients need not be injective. To recover the length of  $C_D/HC_D$  from the displayed formula, we control the kernels of  $V_q \rightarrow W_q$  by a single trace denominator. There is a nonzero  $g \in A$  with

$$gC_D \subseteq B_1, \quad g^{1/q}C_D \subseteq B_q. \quad (2.7)$$

Indeed, choose a  $K$ -basis  $b_1, \dots, b_r$  of  $L$  in  $D$  and its trace-dual basis  $\beta_1, \dots, \beta_r$ . A common denominator  $g \in A \setminus \{0\}$  makes every  $g\beta_j$  lie in  $D$ . The field  $LK'$  is perfect and equals  $L_{\text{perf}}$ . The ring  $\Lambda$  is normal, since an integral equation and the fractions occurring in it descend to a normal stage  $A^{1/q}$ . For  $c \in C_D$ , each  $\text{Tr}_{LK'/K'}(g\beta_j c)$  is integral over  $\Lambda$  and belongs to  $K'$ , hence belongs to  $\Lambda$ . The trace-dual expansion of  $gc$  proves the first inclusion in (2.7). Taking  $q$ -th roots proves the second.

The kernel of  $V_q \rightarrow W_q$  is killed by  $g^{1/q}$ , by (2.7). Kernel and cokernel of an endomorphism of a finite-length object have equal length. Applied to multiplication by  $g^{1/q}$ , this yields

$$\lambda_\infty \ker(V_q \rightarrow W_q) \leq q^{-h} \text{length}_D D/(H^{[q]}, g) = O(q^{-1}).$$

For the estimate, if  $H$  has  $s$  generators then  $H^{s(q-1)+1} \subseteq H^{[q]}$ . Thus  $H^{[q]}$  contains a power of  $\mathfrak{d}$  with exponent  $O(q)$ , whereas  $\dim D/(g) \leq h-1$  if  $g$  is a nonunit. Hilbert–Samuel growth gives the asserted bound; for a unit  $g$  the quotient is zero. The  $W_q$  increase to  $C_D/HC_D$  and are uniformly primary-torsion. The Hilbert–Kunz limit and Lemma 2.4 show

$$\lambda_D(C_D/HC_D) = e_{\text{HK}}(H, D). \quad (2.8)$$

Here we use Monsky’s existence theorem for Hilbert–Kunz multiplicity [20]; see also [19, Section 2.2].

We have identified primary-quotient lengths with Hilbert–Kunz multiplicities. For a parameter ideal, it remains to obtain the Hilbert–Samuel value and zero positive Koszul homology. If  $z$  is a parameter system of  $D$ , finite-stage flatness gives

$$\lambda_\infty H_i(z; B_q) = q^{-h} \text{length}_D H_i(z_1^q, \dots, z_h^q; D).$$

The Koszul complex  $K(z; D)$  has finite free terms in degrees  $0, \dots, h$  and finite-length homology. The Frobenius homology theorem of Roberts, in the form [19, Theorem 2.4], says that the displayed quantity tends to zero for  $i > 0$ ; the original result is [24]. Its hypotheses require precisely a complete characteristic- $p$  ring and a finite-free complex of length its dimension with finite-length homology. Filtered colimits commute with homology, and these Koszul homology modules have the common

annihilator  $(z)$ . Lemma 2.4 proves (2.4), also with  $z$  replaced by  $z^v$ . Finally the Koszul Euler characteristic formula gives

$$\sum_{i=0}^h (-1)^i \text{length}_D H_i(z_1^q, \dots, z_h^q; D) = q^h e((z), D).$$

After division by  $q^h$ , the positive-degree terms tend to zero. Together with (2.8) and the multiplicity power rule this proves (2.5).

Now suppose  $D$  has mixed characteristic. Let  $A_\infty$  be the  $p$ -adic completion of  $A$  and define

$$C_D = (D \otimes_A A_\infty)_{\text{perfd}},$$

the perfectoidization of [6, Notation 4.0.1], built on Bhatt–Scholze’s construction [4, Proposition 8.5]. The stated hypotheses there are exactly that  $D$  is a complete mixed-characteristic local domain with perfect residue field and the chosen finite Witt power-series base  $A$ . Choose  $g \in A \setminus \{0\}$  so that  $A[1/g] \rightarrow D[1/g]$  is finite etale. Corollary 4.0.4 of [6] annihilates positive Koszul homology for every parameter system of  $A_n \otimes_A D$  by the almost ideal  $(g)_{\text{perfd}}$ . At  $n = 0$  this includes every parameter system of  $D$ . Proposition 4.0.10 of that paper makes these uniformly  $\mathfrak{m}_A$ -power-torsion homology modules have normalized length zero. Proposition 4.0.14, again at  $n = 0$ , identifies the length of the parameter quotient with its Hilbert–Samuel multiplicity. These two results, applied also to  $z^v$ , prove (2.4) and (2.5) for this single algebra  $C_D$ . Lemma 2.3 permits restriction to the uncompleted tower throughout.

In either characteristic, a module killed by a power of  $\mathfrak{d}$  is killed by a power of  $\mathfrak{m}_A$ , so all claimed lengths are defined. Additivity and the tower properties are those already established. Lastly the multiplicity rank formula, and equality of the residue fields, give  $e((t), D) = \text{rank}_A D e((t), A) = \text{rank}_A D$ . This proves (2.6).  $\square$

The base-parameter case extends to every finite-length module over  $A$ . The same parameter-quotient positivity also prevents the residue quotient of  $C_D$  from having length zero; this will supply the positive residue-length factor in the final degree-zero estimate.

**Corollary 2.6** (Finite-length base change). *In Theorem 2.5, put  $r = \text{rank}_A D$ . For every finite-length  $A$ -module  $V$ ,*

$$\lambda_D \text{Tor}_i^A(V, C_D) = 0 \quad (i > 0), \quad \lambda_D(V \otimes_A C_D) = r \text{length}_A(V).$$

Moreover  $\lambda_D(C_D/\mathfrak{d}C_D) > 0$ .

*Proof.* For  $V = k$ , the first two assertions are (2.4) and (2.6), using the Koszul resolution over regular  $A$ . Induct on a composition series of  $V$ . The Tor long exact sequences and length additivity discard all positive-degree terms and show that degree-zero length increases by  $r$  at each step. All modules here have bounded primary torsion.

For the last assertion choose a parameter ideal  $H$  of  $D$ . Tensor a composition series of  $D/H$  with  $C_D$ . Right exactness gives a filtration of  $C_D/H C_D$  whose factors are quotients of  $C_D/\mathfrak{d}C_D$ . Therefore

$$0 < e(H, D) = \lambda_D(C_D/H C_D) \leq \text{length}_D(D/H) \lambda_D(C_D/\mathfrak{d}C_D).$$

This proves the required strict positivity.  $\square$

### 3 Rational descent with closed supports

We compare ordinary and normalized Euler lengths through rational  $K$ -theory with support. In the regular-base applications, the strategy is to compare the two evaluations on classes pulled back from the base, then show that sums of conjugate classes come from rational classes over that base. The latter step is the descent theorem below; the final subsection combines it with the normalized-length input of Section 2. We retain connective associative coefficients because Section 4 applies the same descent theorem to the quotient by negligible-rank sequences.

All derived algebras in this section are indexed homologically. Thus connective means that their homotopy groups vanish in negative degrees.

Let  $A$  be a Noetherian commutative  $\mathbb{Z}_{(p)}$ -algebra of finite Krull dimension, let  $J \subset A$  be an ideal containing  $p$ , and let  $H$  be a connective associative  $A$ -algebra. The image of  $A$  acts centrally. For a commutative  $A$ -algebra  $B$ , write

$$\mathrm{Perf}_J(H \otimes_A^{\mathbf{L}} B)$$

for the category of perfect modules whose central localizations vanish off  $V(JB)$ . Its nonconnective  $K$ -theory spectrum and its zeroth homotopy group are denoted by  $K^J(H \otimes_A^{\mathbf{L}} B)$  and  $K_0^J(H \otimes_A^{\mathbf{L}} B)$ , respectively. Rationalization of a spectrum is indicated by a subscript  $\mathbb{Q}$ . For a finite group action,  $(-)^{hG}$  denotes homotopy fixed points (derived invariants), whereas  $(-)^G$  denotes ordinary invariants.

**Theorem 3.1** (Supported rational descent). *Let  $A$  be a commutative Noetherian  $\mathbb{Z}_{(p)}$ -algebra of finite Krull dimension, let  $J \subset A$  be an ideal containing  $p$ , and let  $H$  be a connective associative  $A$ -algebra with central  $A$ -action. Let  $T$  be a finite commutative  $A$ -algebra with an action of a finite group  $G$  over  $A$ . Suppose that  $\mathrm{Spec} T \rightarrow \mathrm{Spec} A$  is surjective and that  $G$  acts transitively on the points of every geometric fiber. Then pullback induces an equivalence*

$$K^J(H)_{\mathbb{Q}} \simeq (K^J(H \otimes_A^{\mathbf{L}} T)_{\mathbb{Q}})^{hG}. \quad (3.1)$$

In particular, the map

$$K_0^J(H) \otimes \mathbb{Q} \longrightarrow (K_0^J(H \otimes_A^{\mathbf{L}} T) \otimes \mathbb{Q})^G \quad (3.2)$$

is surjective.

Neither flatness nor tame ramification of  $A \rightarrow T$  is required. The coefficient algebra  $H$  need not be commutative, finite over  $A$ , or bounded in positive homological degrees. Its connectivity is essential to the argument.

#### 3.1 The truncating invariant and its supports

We use the following precise form of the results of Land–Tamme. A localizing invariant takes exact sequences of stable categories to fiber sequences. It is called *truncating* if its value on a connective associative algebra is unchanged by passage to  $\pi_0$ . Write  $\mathrm{fib}$  for homotopy fiber and  $\mathrm{HN}(-/\mathbb{Q})$  for negative cyclic homology over  $\mathbb{Q}$ . Rational infinitesimal  $K$ -theory

$$I(U) = \mathrm{fib}(K(U)_{\mathbb{Q}} \longrightarrow \mathrm{HN}(U \otimes \mathbb{Q}/\mathbb{Q}))$$

is truncating by Goodwillie’s relative comparison theorem [12, Main Theorem], with the passage from simplicial rings to connective associative ring spectra explained in [18, proof of Corollary 3.9 and Remark 3.10]. Truncating localizing invariants satisfy  $\mathrm{cdh}$  descent, also after tensoring with a fixed connective associative algebra over the base [18, Theorem A.2 and Appendix B, especially

Theorem B.3]. The cdh topology is generated by Nisnevich covers and abstract blowup squares: a proper map and a closed center, with the proper map an isomorphism off that center.

For a derived  $\mathbb{Z}_{(p)}$ -algebra  $U$ , consider

$$E(U) = \text{fib}(K(U)_{\mathbb{Q}} \longrightarrow K(U[1/p])_{\mathbb{Q}}). \quad (3.3)$$

The negative cyclic homology terms in the two character maps are naturally identical, since  $U \otimes \mathbb{Q} \simeq U[1/p] \otimes \mathbb{Q}$ . Taking fibers of the character square therefore gives

$$E(U) \simeq \text{fib}(I(U) \longrightarrow I(U[1/p])). \quad (3.4)$$

It follows that  $E$  is a truncating localizing invariant of  $\mathbb{Z}_{(p)}$ -linear categories. Notice that this argument does not identify  $K(U)_{\mathbb{Q}}$  with  $K(U \otimes \mathbb{Q})$ .

Tensoring with  $H$  preserves the localizing property. It also preserves truncation: if  $B$  is connective, then  $H \otimes_A^{\mathbf{L}} B$  is connective and

$$\pi_0(H \otimes_A^{\mathbf{L}} B) = \pi_0(H) \otimes_A \pi_0(B).$$

Thus replacing  $B$  by  $\pi_0(B)$  does not change the value of  $E$ . For a finite-type  $A$ -scheme  $Y$ , let  $H_Y$  be the derived pullback of  $H$  and put  $Z_Y = Y \times_A V(J)$ . Define

$$\mathcal{L}(Y) = \text{fib}(E(Y; H) \longrightarrow E(Y \setminus Z_Y; H)). \quad (3.5)$$

Here the coefficient notation means evaluation on the category of perfect  $H_Y$ -modules. The coefficient theorem just cited gives cdh descent for  $E(-; H)$ . Restriction to the open complement of  $Z_Y$  preserves the squares defining this topology, and limits commute with limits. Hence  $\mathcal{L}$  also satisfies cdh descent.

Because  $p \in J$ , the support  $Z_Y$  disappears after inverting  $p$ . Expanding the two fibers in (3.5) and using perfect-complex localization consequently identifies

$$\mathcal{L}(Y) \simeq K(\text{Perf}_{Z_Y}(H_Y))_{\mathbb{Q}}. \quad (3.6)$$

In particular, this is the desired rational supported  $K$ -theory, not a cdh replacement of it. The functor is nilinvariant as well.

We recall why the support condition is compatible with the coefficient categories used here. If  $P$  is perfect and  $z \in A$ , compactness identifies localization of its endomorphism spectrum with the endomorphism spectrum of  $P[1/z]$ . Therefore  $P[1/z] = 0$  if and only if some power  $z^v$  kills  $\text{id}_P$  in the homotopy category. Applying this to finitely many generators of  $J$  gives the central support criterion. The corresponding Koszul tensors generate the compact kernel of restriction to the complement. Perfect localization, including idempotent completion, thus computes the fiber in (3.6).

The geometric localization input is Thomason–Trobaugh’s perfect localization theorem [29, Propositions 5.2.2–5.2.4 and Theorem 7.4]. Tensoring its exact sequence with the perfect coefficient category, using [18, Lemmas 3.12 and B.2] and the preceding supported-kernel identification, gives the coefficient version. Nonconnective  $K$ -theory sends this exact sequence to a fiber sequence by [5, Definition 8.1, Theorems 8.7 and 9.8, and Section 9.3].

## 3.2 Descent for finite surjections

The functor  $\mathcal{L}$  is now a nilinvariant cdh sheaf computing actual rational supported  $K$ -theory. To apply it to the finite map in Theorem 3.1, we need descent for finite surjections, which need not be flat. Rational transfers give the finite flat case; flattening by blowup and induction on closed subsets will give the general case.

**Lemma 3.2.** *The functor  $\mathcal{L}$  satisfies ordinary Čech descent for finite faithfully flat covers, universally under base change.*

*Proof.* Let  $f : X \rightarrow Y$  be finite faithfully flat. Since the schemes are Noetherian,  $f_*\mathcal{O}_X$  is a vector bundle of positive rank. Restriction of scalars preserves perfect coefficient modules and their supports: locally the scalar-extended coefficient algebra is a finite projective module over the original one. It therefore defines a transfer. Finite-flat base change and the projection formula give natural transformations of presheaves on  $Y$ -schemes

$$\mathcal{L} \xrightarrow{f^*} \mathcal{L}_X \xrightarrow{\mathrm{tr}_f} \mathcal{L}, \quad \mathcal{L}_X(W) = \mathcal{L}(W \times_Y X),$$

whose composite is tensoring with the pullback of  $f_*\mathcal{O}_X$ .

Work first on a Zariski open of  $Y$  on which that vector bundle is free of rank  $a > 0$ , and fix a trivialization there. Additivity of  $K$ -theory identifies this composite with multiplication by  $a$ . The same trivialization pulls back on every  $Y$ -scheme, so this identity is natural on the whole presheaf diagram. Rationally,  $a^{-1} \mathrm{tr}_f$  makes  $\mathcal{L}$  a retract of  $\mathcal{L}_X$ .

After base change to  $X$ , the augmented Čech nerve of  $f$  has a section and hence an extra degeneracy. Its image under any presheaf has split descent. Thus  $\mathcal{L}_X$  satisfies descent for  $f$ . The descent comparison for  $\mathcal{L}$  is a retract of this equivalence, and is itself an equivalence. Finally, both its source and its totalization are Zariski sheaves. The conclusion therefore glues over the trivializing cover of  $Y$ . All constructions commute with base change.  $\square$

**Lemma 3.3.** *The functor  $\mathcal{L}$  satisfies ordinary Čech descent for every finite surjection of finite-type  $A$ -schemes.*

*Proof.* Let  $\tau$  be the topology generated by cdh covers and finite faithfully flat covers. These cover families are stable under base change. By Lemma 3.2 and cdh descent,  $\mathcal{L}$  is a  $\tau$ -sheaf. We show that every finite surjection  $X \rightarrow Y$  is a  $\tau$ -cover.

Reductions and finite closed decompositions are cdh-local operations. For clarity, if a reduced scheme is a union  $Y = Y_1 \cup Y_2$  of reduced closed subschemes, the proper map  $Y_1 \rightarrow Y$  is an isomorphism off  $Y_2$  and gives the abstract blowup square with intersection  $Y_1 \cap Y_2$ . This reduces closed unions to their pieces. We may consequently reduce to integral  $Y$ , with induction on dimension for proper closed subsets.

Choose a reduced irreducible component  $X_0$  of  $X$  dominating  $Y$ . The map  $X_0 \rightarrow Y$  is finite and surjective. On a dense open  $U \subset Y$  it is finite flat of positive rank. Flattening by blowup, applied to  $\mathcal{O}_{X_0}$ , gives a blowup  $Y' \rightarrow Y$ , an isomorphism over  $U$ , for which the strict transform  $X'_0 \rightarrow Y'$  is flat and finitely presented [28, Tag 0815]. The strict transform is a closed subscheme of  $X_0 \times_Y Y'$ , so it is finite over  $Y'$ . The blowup  $Y'$  is integral, and the image of  $X'_0$  contains its generic point. Finiteness makes that image closed. Thus  $X'_0 \rightarrow Y'$  is surjective, and is a finite faithfully flat cover factoring through  $X \times_Y Y'$ .

Let  $Z \subsetneq Y$  be the blowup center. The family  $\{Y' \rightarrow Y, Z \rightarrow Y\}$  is a cdh cover. By induction on dimension, the finite surjection  $X \times_Y Z \rightarrow Z$  is a  $\tau$ -cover. Together with  $X'_0 \rightarrow Y'$ , these maps give a  $\tau$ -covering family of  $Y$  that factors through  $X$ . Hence  $X \rightarrow Y$  itself generates a covering sieve. The induction starts in dimension zero, where a reduced integral base is a field and any nonzero finite algebra is finite faithfully flat. This proves the assertion, and therefore the required Čech descent.  $\square$

The argument uses actual retracts and sheaf limits; it requires neither hyperdescent nor convergence of an unbounded descent spectral sequence.

### 3.3 The geometric orbit argument

We have established descent for finite surjections. It remains to use geometric transitivity to identify the homotopy  $G$ -invariants with the value on the base. After arranging a section and reducing the schemes, we compare its translates away from their coincidence loci and use induction along those loci.

*Proof of Theorem 3.1.* Write  $X = \operatorname{Spec} T$  and  $S = \operatorname{Spec} A$ . On finite-type  $S$ -schemes consider the natural comparison

$$\mathcal{L}(Y) \longrightarrow \mathcal{L}(X \times_S Y)^{hG}. \quad (3.7)$$

Both sides are sheaves for the topology used in Lemma 3.3: base change preserves the cover families, and homotopy fixed points commute with limits. Applying finite-surjection descent to  $X \rightarrow S$ , it suffices to prove (3.7) on the nonaugmented terms of its Čech nerve. Over each such term, the pulled-back  $X$ -cover admits a section, given by a projection to one of its  $X$ -coordinates. The geometric transitivity hypothesis is stable under base change. We are therefore reduced to covers admitting a section.

We prove this case by induction on the dimension of  $Y$ . Nilinvariance and the closed-union squares described in the proof of Lemma 3.3 reduce the base to an integral reduced scheme. Indeed both sides of (3.7) have these descent properties; the component reduction uses finitely many closed pieces and lower-dimensional intersections.

Let  $s : Y \rightarrow X \times_S Y$  be a section. Its finitely many  $G$ -translates cover the source on points. To see this, extend the residue field at any point of  $Y$  to an algebraic closure and use transitivity on that geometric fiber. Each translate is a closed immersion, because the source is separated over  $Y$ , and each factors through the reduction

$$V = (X \times_S Y)_{\text{red}}.$$

Retain the distinct translated sections and index them by a transitive finite  $G$ -set  $I$ . Two distinct sections have a proper closed coincidence locus in  $Y$ : their equalizer is closed, and agreement at the generic point of integral reduced  $Y$  would imply agreement everywhere. Let  $Z \subsetneq Y$  be the reduced union of these coincidence loci, with  $Z = \emptyset$  if there is only one section.

The map

$$q : W = \coprod_{i \in I} Y \longrightarrow V$$

is finite and surjective. Over  $Y \setminus Z$  the section images are pairwise disjoint and cover the reduced scheme  $V$ . Their disjoint union is therefore isomorphic to  $V$  there. Consequently  $q$  is a proper modification and the square

$$\begin{array}{ccc} W \times_Y Z & \longrightarrow & W \\ \downarrow & & \downarrow \\ V \times_Y Z & \longrightarrow & V \end{array}$$

is an abstract blowup square. It is  $G$ -equivariant.

The  $G$ -action on  $W$  merely permutes its copies of  $Y$ . The canonical restriction maps give a  $G$ -equivariant equivalence  $\mathcal{L}(W) \simeq \prod_{i \in I} \mathcal{L}(Y)$ . A stabilizer of one section acts identically on that copy, including on the coefficient algebra pulled back from  $Y$ . This is a canonical identity of coefficient-bearing  $Y$ -schemes, so the action is the ordinary permutation action, with coherently trivial stabilizer action. The same statement holds after restriction to  $Z$ . Since  $\mathcal{L}$  is rational, averaging over a finite group is exact. It follows that

$$\mathcal{L}(W)^{hG} \simeq \mathcal{L}(Y), \quad \mathcal{L}(W \times_Y Z)^{hG} \simeq \mathcal{L}(Z). \quad (3.8)$$

For example, if  $I = G/G_s$ , the first fixed-point spectrum is  $\mathcal{L}(Y)^{hG_s}$  for the trivial  $G_s$ -action, and is  $\mathcal{L}(Y)$ . This identity holds without boundedness: the trivial  $\mathbb{Q}[G_s]$ -module  $\mathbb{Q}$  is projective, so derived invariants are ordinary invariants even on unbounded rational complexes.

Apply  $\mathcal{L}$  to the abstract blowup square and take homotopy  $G$ -invariants. The inductive hypothesis over  $Z$  identifies

$$\mathcal{L}(V \times_Y Z)^{hG} \simeq \mathcal{L}(Z).$$

Here nilinvariance allows one to pass between  $V \times_Y Z$  and the corresponding reduced or unreduced pullback of the original cover. Combining this with (3.8), the pullback square gives

$$\mathcal{L}(V)^{hG} \simeq \mathcal{L}(Y) \times_{\mathcal{L}(Z)} \mathcal{L}(Z) \simeq \mathcal{L}(Y).$$

Nilinvariance restores  $X \times_S Y$  in place of  $V$ . In dimension zero there are no proper nonempty coincidence loci, and the same proof is simply the calculation for a disjoint union of section copies. This starts the induction and proves (3.7).

Taking  $Y = S$  and using (3.6) yields (3.1). Finally, exact rational invariants give

$$\pi_0\left((K^J(H \otimes_A^{\mathbf{L}} T)_{\mathbb{Q}})^{hG}\right) = (K_0^J(H \otimes_A^{\mathbf{L}} T) \otimes \mathbb{Q})^G,$$

which proves (3.2), in fact as an isomorphism.  $\square$

### 3.4 Comparison of ordinary and normalized Euler lengths

We now isolate the consequence needed before introducing asymptotic categories. Let  $(A, \mathfrak{m}_A, k)$  be a complete regular local  $\mathbb{Z}_{(p)}$ -algebra of dimension  $e > 0$ , with residue characteristic  $p$ , and let  $z_1, \dots, z_e$  be regular parameters. Let  $T$  be a finite local domain extension of  $A$  with the same residue field. Put  $r = \text{rank}_A T$ . Suppose that  $C$  is a  $T$ -algebra equipped with an additive nonnegative length  $\lambda$  on  $C$ -modules annihilated by powers of  $\mathfrak{m}_A$ , with  $\lambda(0) = 0$ . Infinite length is allowed for general modules. Assume that, for every  $v \geq 1$ ,

$$\lambda H_i(z_1^v, \dots, z_e^v; C) = 0 \quad (i > 0), \quad \lambda(C/(z_1^v, \dots, z_e^v)C) = v^e r. \quad (3.9)$$

Theorem 2.5 supplies exactly these hypotheses in our applications, since the multiplicity of the base parameters on  $T$  is its generic rank over the regular base. The next lemma is the formal version of Corollary 2.6, using only (3.9).

**Lemma 3.4.** *For every finite-length  $A$ -module  $V$ ,*

$$\lambda \text{Tor}_i^A(V, C) = 0 \quad (i > 0), \quad \lambda(V \otimes_A C) = r \text{length}_A(V). \quad (3.10)$$

*All these lengths are finite.*

*Proof.* The Koszul resolution on  $z_1, \dots, z_e$  resolves  $k$  over  $A$ , so (3.9) with  $v = 1$  proves the assertion for  $V = k$ . Induct on the length of  $V$  using a composition series. In the long exact Tor sequence associated to a short exact sequence with quotient  $k$ , nonnegativity and additivity show that every positive Tor term has length zero. The image of the connecting map into degree zero also has length zero. Additivity in degree zero then gives the asserted factor  $r$ . This simultaneously proves finiteness at every step.  $\square$

If a finite group  $G$  acts on  $T$  over  $A$ , then for  $g \in G$  scalar twisting means  $g^*P = P \otimes_{T,g} T$ , where the left  $T$ -action on the last factor is through  $g$ . On a free complex this applies  $g$  to every differential matrix entry.

**Lemma 3.5** (Ordinary norm comparison). *Assume, in addition, that a finite group  $G$  acts on  $T$  over  $A$  and is transitive on every geometric fiber of  $\mathrm{Spec} T \rightarrow \mathrm{Spec} A$ . For a perfect  $T$ -complex  $P$  supported on  $V(\mathfrak{m}_A T)$ , the quantities*

$$\chi_T(P) = \sum_i (-1)^i \mathrm{length}_T H_i(P), \quad \chi_C(P) = \sum_i (-1)^i \lambda H_i(P \otimes_T^{\mathbf{L}} C)$$

*are finite and define homomorphisms on  $K_0^{\mathrm{m}A}(T)$ . They agree on every norm sum:*

$$\sum_{g \in G} \chi_T(g^* P) = \sum_{g \in G} \chi_C(g^* P). \quad (3.11)$$

*More generally, the two homomorphisms agree on the image of  $K_0^{\mathrm{m}A}(A)$  and on every  $G$ -invariant rational class.*

*Proof.* We first establish finiteness and additivity, then compare the evaluations on pullbacks before applying descent to the norm class. Represent  $P$  by a bounded finite free  $T$ -complex. Its ordinary homology is finite and supported at the closed point, hence has finite length. For the  $C$ -evaluation, choose a common  $v$  such that each  $z_j^v$  acts nullhomotopically on  $P$ . This is possible by the support criterion above. Set  $K_v = K(z_1^v, \dots, z_e^v; T)$ . Successively splitting the cones of these nullhomotopic actions gives

$$K_v \otimes_T P \otimes_T C \simeq \bigoplus_{j=0}^e (P \otimes_T C)[j]^{\oplus \binom{e}{j}}. \quad (3.12)$$

The homotopies may be tensored with  $C$  and with the other Koszul factors, so this splitting respects the module structure on which  $\lambda$  is defined.

In the opposite order, taking Koszul homology first gives a finite spectral sequence whose terms are finite direct sums of  $H_j(z_1^v, \dots, z_e^v; C)$ . Their lengths are finite by (3.9). Hence every homology module on the left side of (3.12) has finite length. The splitting and nonnegativity then give finiteness for every  $H_i(P \otimes_T C)$ . The homology modules and spectral-sequence terms to which  $\lambda$  is applied are annihilated by fixed powers of  $\mathfrak{m}_A$ . The Euler sums are therefore defined, and the long exact homology sequence of a triangle proves their additivity.

Let  $Q$  now be a perfect  $A$ -complex supported at  $\mathfrak{m}_A$ . Its homology modules have finite length. The finite hyper-Tor spectral sequence for  $Q \otimes_A^{\mathbf{L}} C$ , together with Lemma 3.4, gives

$$\chi_C(Q \otimes_A^{\mathbf{L}} T) = r \sum_j (-1)^j \mathrm{length}_A H_j(Q). \quad (3.13)$$

Indeed every positive Tor row has length zero and the zeroth row has the displayed scaled lengths.

For the ordinary evaluation, choose a bounded finite free  $A$ -resolution  $L_\bullet$  of  $T$ , possible because  $A$  is regular. Its alternating rank is  $r$ , as is seen after tensoring with the fraction field of  $A$ . For every finite-length  $A$ -module  $V$ , the finite complex  $V \otimes_A L_\bullet$  therefore gives

$$\sum_i (-1)^i \mathrm{length}_A \mathrm{Tor}_i^A(V, T) = \sum_i (-1)^i \mathrm{rank}_A(L_i) \mathrm{length}_A(V) = r \mathrm{length}_A(V).$$

The hyper-Tor spectral sequence for  $Q \otimes_A^{\mathbf{L}} T$  consequently yields

$$\chi_T(Q \otimes_A^{\mathbf{L}} T) = r \sum_j (-1)^j \mathrm{length}_A H_j(Q). \quad (3.14)$$

We have used that finite-length  $T$ -modules have the same  $A$ -length and  $T$ -length, because both residue fields are  $k$ . Equations (3.13) and (3.14) prove agreement on classes pulled back from  $A$ .

Apply Theorem 3.1 with  $H = A$  and  $J = \mathfrak{m}_A$ . Every invariant class in  $K_0^{\mathfrak{m}_A}(T) \otimes \mathbb{Q}$  is the pullback of a rational class downstairs. Extending the two Euler homomorphisms  $\mathbb{Q}$ -linearly, they therefore agree on that invariant subspace. The class  $\sum_{g \in G} [g^* P]$  is invariant, which gives (3.11).  $\square$

*Remark 3.6.* No invariance of  $\lambda$  under  $G$  is asserted or needed. The equality is for the sum of the conjugate classes, not for each individual class. This distinction allows the two length algebras used later to be chosen independently.

## 4 Supported asymptotic complexes

The goal is an analogue of Lemma 3.5 for sequences of bounded free complexes modulo free sequences of negligible normalized rank, proved in Theorem 4.7. Two points require care: support may arise only after passing to the quotient, and supported objects may be idempotent summands whose projectors do not lift before quotienting. Their homology lengths therefore cannot in general be computed from the raw homologies of arbitrary lifts. We first construct nonnegative lengths on uniformly supported lifts, extend them through the supported quotient and its idempotent completion, and prove their compatibility with scalar extension on every supported summand.

Throughout this section,  $(A, J)$  is an excellent regular local  $\mathbb{Z}_{(p)}$ -algebra of dimension  $e > 0$  with residue characteristic  $p$ , where  $J$  is the maximal ideal, and  $z_1, \dots, z_e$  is a regular system of parameters. Let  $T$  be a finite local domain over  $A$  with the same residue field, and put  $r = \text{rank}_A T$ . Let  $C$  be a  $T$ -algebra equipped with a nonnegative additive length  $\lambda$  on its modules annihilated by powers of  $J$ . The assumptions on this length that are used in this section are

$$\lambda H_j(z_1^v, \dots, z_e^v; C) = 0 \quad (j > 0), \quad \lambda(C/(z_1^v, \dots, z_e^v)C) = v^e r \quad (v \geq 1). \quad (4.1)$$

The length is allowed to vanish on nonzero modules. In the applications, Theorem 2.5 supplies these hypotheses, as well as the stronger parameter properties used in the other sections. Write  $K_v$  for the Koszul complex  $K(z_1^v, \dots, z_e^v; A)$ , in homological degrees  $0, \dots, e$ .

### 4.1 The rank quotient and its support condition

Fix positive integers  $s_n$  and a nonprincipal ultrafilter  $\mathcal{U}$  on the positive integers. Rank estimates  $O(s_n)$  and  $o(s_n)$  always refer to the ordinary limit, not to  $\mathcal{U}$ . If  $(a_n/s_n)$  is bounded, write

$$\bar{a} = \lim_{n \rightarrow \mathcal{U}} \frac{a_n}{s_n}.$$

For  $S = A$  or  $T$ , consider the additive category of sequences  $(S^{a_n})_n$  with  $a_n = O(s_n)$ , whose morphisms are arbitrary sequences of matrices over  $S$ . Let  $\mathcal{D}_S$  be its perfect hull: take complexes in one common bounded degree range for the whole sequence, and then idempotent completion. Let  $\mathcal{N}_S$  be the thick subcategory generated by the free sequences with  $a_n = o(s_n)$ , and define

$$\mathcal{P}_S = (\mathcal{D}_S / \mathcal{N}_S)^\natural.$$

Here  $(-)^{\natural}$  denotes idempotent completion. The quotient functor is denoted by  $q_S$ .

A thick subcategory is closed under shifts, cones, and direct summands. The Verdier quotient inverts maps whose cones lie in that subcategory; idempotent completion then adjoins the images of idempotent endomorphisms.

An object  $X$  in  $\mathcal{D}_S$  or  $\mathcal{P}_S$  is called *supported on  $J$*  if there is an integer  $v \geq 1$  such that  $z_j^v \text{id}_X = 0$  for every  $j$ . The identity is taken in the category containing  $X$ . Thus in  $\mathcal{D}_S$  this is a uniform

homotopical support condition on the sequence, whereas in  $\mathcal{P}_S$  the actions need vanish only after the rank-small quotient. Write  $\mathcal{S}_S \subseteq \mathcal{D}_S$  and  $\mathcal{P}_S^J \subseteq \mathcal{P}_S$  for the resulting full subcategories. They are thick: nilpotence of central actions is preserved by shifts, cones, and retracts, allowing larger powers for cones. We identify this condition with central-localization support before applying descent at the end of the section.

*Remark 4.1.* Raw homology lengths need not survive the rank quotient. Over a discrete valuation ring  $A = T$  with parameter  $z$ , take  $s_n = n$  and the rank-one free resolution of  $T/(z^n)$ . This sequence belongs to  $\mathcal{N}_T$ , whereas its raw normalized degree-zero homology length is 1. It has no uniform annihilating power before quotienting. For each fixed  $v$ , however, its homology lengths after tensoring with  $K(z^v; T)$  are bounded independently of  $n$  and hence become zero after normalization. This is why we construct homology lengths on uniformly supported lifts first, and then extend them to supported quotient objects.

## 4.2 The supported subquotient

The next lemma supplies that extension's categorical starting point: every supported quotient object is a summand of a uniformly supported lift, and morphisms between such lifts can be computed using supported roofs. Neither assertion requires the summand projector to lift.

**Lemma 4.2.** *The natural exact functor*

$$\mathcal{S}_S/(\mathcal{S}_S \cap \mathcal{N}_S) \longrightarrow \mathcal{D}_S/\mathcal{N}_S$$

*is fully faithful. Its idempotent completion is  $\mathcal{P}_S^J$ .*

*Proof.* First let  $X \in \mathcal{S}_S$ ,  $Y \in \mathcal{N}_S$ , and  $f : X \rightarrow Y$ . Choose  $v$  such that  $z_j^v \text{id}_X = 0$  for every  $j$ . Centrality gives  $z_j^v f = 0$ . The exact Hom sequence of the triangle

$$\text{fib}(z_j^v : Y \rightarrow Y) \longrightarrow Y \xrightarrow{z_j^v} Y$$

therefore lifts  $f$  through the first object. Repeat this operation for all the parameters, always using the zero action on the original source  $X$ . The result factors  $f$  through  $K_v \otimes_A Y[-e]$ . This object belongs to  $\mathcal{N}_S$  by thickness, and to  $\mathcal{S}_S$  by the usual Koszul homotopies. Thus every map from a supported object to a small object factors through a supported small object.

Here is the resulting roof argument. A morphism between  $X, Y$  in  $\mathcal{S}_S$ , computed in  $\mathcal{D}_S/\mathcal{N}_S$ , has a representative

$$X \xleftarrow{s} Z \xrightarrow{f} Y, \quad \text{cofib}(s) = N_0 \in \mathcal{N}_S.$$

In the triangle  $Z \rightarrow X \xrightarrow{u} N_0$ , factor  $u$  through  $N_1 \in \mathcal{S}_S \cap \mathcal{N}_S$  by the preceding paragraph. Let  $Z_1$  be the fiber of  $X \rightarrow N_1$ . The induced morphism of triangles gives  $c : Z_1 \rightarrow Z$  and  $s_1 : Z_1 \rightarrow X$  with  $sc = s_1$ . Now  $Z_1 \in \mathcal{S}_S$  and  $\text{cofib}(s_1) = N_1$ , so  $X \xleftarrow{s_1} Z_1 \xrightarrow{fc} Y$  is a representative entirely in the supported subquotient. This proves fullness.

For faithfulness, suppose a map  $f : Z \rightarrow Y$  between supported objects becomes zero in  $\mathcal{D}_S/\mathcal{N}_S$ . The calculus of Verdier roofs gives a map  $t : W \rightarrow Z$  with small cone and  $ft = 0$ . The triangle of  $t$  makes  $f$  factor through that small cone. Factoring its first leg through a supported small object as above shows that  $f$  is already zero in the supported subquotient. Apply this to the numerator of any roof to obtain faithfulness.

Finally take  $X \in \mathcal{P}_S^J$ . It is a retract of  $q_S(Y)$  for some  $Y \in \mathcal{D}_S$ . Choose  $v$  with all  $z_j^v$  acting as zero on  $X$ . The cone of each such zero map splits, so  $X$  is a retract of  $K_v \otimes_A X$ . The latter is a retract of  $q_S(K_v \otimes_A Y)$ , whose lift is uniformly supported. Hence  $X$  belongs to the idempotent completion of the fully faithful supported subquotient. The reverse inclusion follows because support is closed under retracts.  $\square$

### 4.3 Nonnegative homology lengths

Every supported quotient object is now accessible through a uniformly supported lift and a summand. We next construct its nonnegative homology lengths and show that they give additive Euler evaluations.

For ordinary length, let  $\mathcal{A}_T$  be the abelian category of sequences of finite  $T$ -modules annihilated by some common power of  $J$  and with lengths  $O(s_n)$ . For the  $C$  evaluation, let  $\mathcal{A}_C$  be the category of sequences of  $C$ -modules with the same uniform torsion condition and with finite  $\lambda$ -lengths  $O(s_n)$ . The power and the implied constant can depend on the object. Kernels, cokernels, and extensions preserve these conditions, so both categories are abelian. In each case the sequences of normalized ultralimit length zero form a Serre subcategory. Denote the Serre quotients by  $\overline{\mathcal{A}}_T$  and  $\overline{\mathcal{A}}_C$ , and their nonnegative additive lengths by  $\overline{\text{length}}_T$  and  $\overline{\lambda}$ , respectively. These lengths are finite on every object. Here negligible length means zero ultralimit; the free sequences generating  $\mathcal{N}_T$  still satisfy the stronger ordinary estimate  $o(s_n)$ . We will show that supported objects killed by the rank quotient have zero homology in these abelian quotients.

**Proposition 4.3.** *There are homological functors*

$$H_i^{\text{ord}} : \mathcal{P}_T^J \longrightarrow \overline{\mathcal{A}}_T, \quad H_i^C : \mathcal{P}_T^J \longrightarrow \overline{\mathcal{A}}_C$$

*whose values on a uniformly supported lift are its ordinary homology sequence and its homology sequence after tensoring with  $C$ , respectively. For each object  $X$ , their lengths*

$$b_i^{\text{ord}}(X) = \overline{\text{length}}_T H_i^{\text{ord}}(X), \quad b_i^C(X) = \overline{\lambda} H_i^C(X)$$

*are finite nonnegative real numbers, zero outside a finite range. Their Euler sums define homomorphisms*

$$\chi_{\text{ord}}, \chi_C : K_0(\mathcal{P}_T^J) \longrightarrow \mathbb{R}.$$

*If  $F = (F_n)$  is a bounded free sequence whose image is supported, and  $v$  makes all  $z_j^v$  act as zero on that image, then for either choice of evaluation, writing  $b_i(F) = b_i(q_T F)$ ,*

$$(1+t)^e \sum_i b_i(F) t^i = \sum_i t^i \lim_{n \rightarrow \mathcal{U}} \frac{\text{len } H_i(K_v \otimes_A F_n)}{s_n}. \quad (4.2)$$

*Here  $\text{len} = \text{length}_T$  in the ordinary case; in the  $C$  case the complex on the right is further tensored over  $T$  with  $C$ , and  $\text{len} = \lambda$ . The identity is one of finite Laurent polynomials.*

*Proof.* We first bound homology on  $\mathcal{S}_T$ , then show that it kills  $\mathcal{S}_T \cap \mathcal{N}_T$ , and finally extend it to the idempotent-completed quotient.

Begin with a bounded free sequence  $F$  in  $\mathcal{S}_T$ . Choose uniform powers annihilating its identity. Splitting their cones gives an isomorphism in the homotopy category

$$K_v \otimes_A F \simeq \bigoplus_{j=0}^e F[j]^{\oplus \binom{e}{j}}. \quad (4.3)$$

For ordinary length, every Koszul homology  $H_j(K_v \otimes_A T)$  has finite length. Taking Koszul homology first in the bounded double complex bounds the homology lengths of  $K_v \otimes_A F_n$  by fixed constants times the ranks of the terms of  $F_n$ . They are therefore  $O(s_n)$ . For  $C$ , the same spectral sequence uses Equation (4.1): positive Koszul rows have length zero and the zero row has finite length. Additivity and nonnegativity give the same bound. The splitting in Equation (4.3) bounds the

homologies of  $F_n$  themselves. They have a common primary annihilator, because a power of every  $z_j$  annihilates them.

A general object  $X \in \mathcal{S}_T$  may be a retract in  $\mathcal{D}_T$  of a bounded free sequence  $F$  that is not supported. Choose  $v$  killing the parameter actions on  $X$ . Then  $X$  is a retract of  $K_v \otimes_A X$ , which is a retract of the uniformly supported bounded free sequence  $K_v \otimes_A F$ . The preceding bounds apply to this sequence and hence to the homology summands defining  $X$ . This is a retract before taking the rank quotient. Thus homology on  $\mathcal{S}_T$  takes values in the indicated abelian length quotients.

Fix  $v$ . Koszul tensoring a rank-small free generator has homology of normalized length zero by the same bounds. The class of objects  $Y \in \mathcal{D}_T$  with this property for  $K_v \otimes_A Y$  is thick: use homology long exact sequences, nonnegativity, and retracts. It therefore contains  $\mathcal{N}_T$ . If  $X \in \mathcal{S}_T \cap \mathcal{N}_T$ , choose  $v$  killing its parameter actions. Then  $X$  is a retract of  $K_v \otimes_A X$ , so its homology is zero in both length quotients.

The homology functors consequently invert maps whose cones are supported and small. They descend to  $\mathcal{S}_T/(\mathcal{S}_T \cap \mathcal{N}_T)$ , preserving homology long exact sequences. Now consider summands after quotienting. By Lemma 4.2, every object of  $\mathcal{P}_T^J$  is a summand of an object of this subcategory. Extend a homology functor to a summand  $(X, a)$  by taking  $\text{im } H_i(a)$  in its abelian target. This does not require a sequence of chain projectors lifting  $a$ . Distinguished triangles in an idempotent completion are summands of distinguished triangles in the original triangulated category [1, Definition 1.10 and Theorem 1.12]. Taking images of compatible idempotents preserves exact sequences in an abelian category, so the extended functors are still homological.

Each object is a summand of a fixed bounded uniformly supported Koszul lift. Its homology objects therefore vanish outside a finite range, and their lengths are bounded by those of that lift. This proves finiteness, nonnegativity, and boundedness of the  $b_i$ . The finite long exact sequences prove additivity of the Euler sums.

For the last assertion, the splitting in Equation (4.3) now takes place in  $\mathcal{P}_T$ , since the chosen powers act as zero there. Applying the extended homological functors gives the left side of Equation (4.2). Its Koszul tensor has a uniformly supported lift, whose actual homology computes the right side. This proves the identity without defining its coefficients by formal division.  $\square$

In particular, Equation (4.2) assigns zero to the quotient object in Remark 4.1, as required. Its nonnegative coefficients come from homological functors, not from an assumed positivity of a polynomial quotient.

#### 4.4 Scalar extension and nonlifting idempotents

Construct  $\overline{\mathcal{A}}_A$  and its length in the same way using ordinary finite  $A$ -modules. The preceding construction over  $A$  with ordinary length defines  $\chi_A$  on  $\mathcal{P}_A^J$ . Scalar extension gives an exact functor  $\mathcal{P}_A^J \rightarrow \mathcal{P}_T^J$ .

The supported Euler evaluations are defined on every object of  $\mathcal{P}_T^J$ . To apply descent, it remains to prove that they agree on all classes coming from  $\mathcal{P}_A^J$ , including idempotent summands.

The normalized evaluation uses the following consequence of the length hypotheses. We state it for the present base  $A$ , which need not be complete.

**Lemma 4.4.** *For every finite-length  $A$ -module  $V$ ,*

$$\lambda \text{Tor}_j^A(V, C) = 0 \quad (j > 0), \quad \lambda(V \otimes_A C) = r \text{length}_A V.$$

*All these modules are annihilated by a power of  $J$ .*

*Proof.* The Koszul and composition-series proof of Lemma 3.4 applies here: it uses regularity and the length hypothesis, but not completeness. A power of  $J$  annihilating  $V$  also annihilates its derived tensor homology.  $\square$

**Proposition 4.5.** *For every  $X \in \mathcal{P}_A^J$ ,*

$$\chi_{\text{ord}}(X \otimes_A^{\mathbf{L}} T) = r\chi_A(X) = \chi_C(X \otimes_A^{\mathbf{L}} T).$$

*These equalities hold for summands defined by idempotents in the Verdier quotient, without assuming that the idempotents lift to chain maps before localization.*

*Proof.* The two evaluations recover the factor  $r$  differently: the ordinary one uses the alternating rank of a resolution of  $T$ , while the normalized one uses the vanishing of positive Tor lengths. To make both calculations on an arbitrary summand, we descend their finite spectral sequences through supported Verdier roofs before taking idempotent images.

Let  $Y$  be a uniformly supported bounded free sequence over  $A$  and let  $a$  be an idempotent endomorphism of its image in  $\mathcal{S}_A/(\mathcal{S}_A \cap \mathcal{N}_A)$ . Every supported object of  $\mathcal{P}_A$  has this form as a summand, by Lemma 4.2: in its proof one may enlarge the initial lift to a bounded free sequence before taking the Koszul tensor. Put

$$V_b = \text{im } H_b^A(a) \quad \text{in } \overline{\mathcal{A}}_A.$$

These images exist in the abelian quotient even when  $a$  has no chain-level projector representing it.

Choose a fixed bounded finite free  $A$ -resolution  $P_\bullet$  of  $T$ ; such a resolution exists because  $A$  is regular. The double complex  $Y \otimes_A P_\bullet$  computes  $Y \otimes_A^{\mathbf{L}} T$ . Filtering by the degree of  $P$  gives a finite spectral sequence with first page

$$E_{u,b}^1 = H_b(Y) \otimes_A P_u. \tag{4.4}$$

It is natural in homotopy-category maps from this page onward: homotopies in  $Y$  preserve the filtration and act trivially on the first page. On retracts in the prequotient category the same construction is defined by taking the corresponding idempotent images on every page and on the abutment filtration. Its terms and finite abutment filtration can be regarded as  $A$ -modules. The abutment's ordinary  $A$ -length equals its  $T$ -length, since the residue fields agree.

A map with supported small cone is an isomorphism on  $H_b(Y)$  in  $\overline{\mathcal{A}}_A$ . It is thus an isomorphism on Equation (4.4), on every subsequent page, and on the finite filtered abutment in the length quotient. The spectral sequence therefore descends through all Verdier roofs in the supported subquotient. The natural augmentation  $\text{Tot}(Y \otimes_A P_\bullet) \rightarrow Y \otimes_A T$  identifies its abutment with the homology of scalar extension, and this identification also descends through those roofs. Thus the induced action of  $a$  on the abutment is precisely the scalar-extended idempotent, not a separately chosen action.

In particular  $a$  acts as an actual idempotent on every page and on that filtered abutment. The differentials commute with this action. Taking its images commutes with passage to homology, since an idempotent decomposes a complex in an abelian category into its image and complementary image. It also preserves the finite exact sequences defining the abutment filtration: for a filtration preserved by an idempotent  $a$ , one has  $\text{im}(a) \cap F_u = a(F_u)$  and hence  $\text{gr}_u \text{im}(a) = \text{im}(\text{gr}_u a)$ . Consequently the Euler length on the scalar-extended summand is the Euler length of the images on the first page. They are  $\text{rank}_A(P_u)$  copies of  $V_b$ , and hence this value is

$$\left( \sum_u (-1)^u \text{rank}_A P_u \right) \sum_b (-1)^b \overline{\text{length}}_A V_b = r \sum_b (-1)^b \overline{\text{length}}_A V_b. \tag{4.5}$$

The equality of alternating ranks with  $r$  follows by tensoring the resolution with the fraction field of  $A$ .

For  $C$ , use the finite hyper-Tor spectral sequence

$${}^C E_{u,b}^2 = \mathrm{Tor}_u^A(H_b(Y), C) \implies H_{u+b}(Y \otimes_A^{\mathbf{L}} C). \quad (4.6)$$

It can be constructed from the functorial finite Postnikov filtration of  $Y$ . It has bounded  $b$  and  $0 \leq u \leq e$ , so there is no unbounded convergence issue. Lemma 4.4 makes every positive Tor term zero in  $\overline{\mathcal{A}}_C$ . It also implies that  $V \mapsto V \otimes_A C$  induces an exact functor

$$\overline{\mathcal{A}}_A \longrightarrow \overline{\mathcal{A}}_C$$

that multiplies length by  $r$ . Indeed, on a short exact sequence its possible failure of left exactness is the image of a positive Tor module of zero length. Negligible-length sequences map to negligible-length sequences by the same lemma. This explains both exactness after the Serre quotient and descent of the functor to that quotient.

It follows that a supported-small-cone map induces isomorphisms on the second page of Equation (4.6) and its finite filtered abutment in the length quotient. Thus this spectral sequence too is functorial through the supported Verdier roofs. Taking the images of the same idempotent  $a$  gives only the zero Tor row, namely the images of  $V_b$  under this exact tensor functor. Their Euler length is  $r \sum_b (-1)^b \overline{\mathrm{length}}_A V_b$ , exactly the value in Equation (4.5). Finally this last sum is  $\chi_A(Y, a)$  by construction. This proves both equalities for every summand and hence for every  $X \in \mathcal{P}_A^J$ .  $\square$

#### 4.5 Connective coefficients and the norm comparison

We have constructed the supported Euler evaluations and proved their agreement after scalar extension from  $A$ . To extend that agreement to conjugate sums, we now put the rank quotient in the form required by Theorem 3.1. The coefficient algebra must be connective, and the support condition must be central-localization support.

**Lemma 4.6** (Connective coefficients). *There is a connective associative derived  $A$ -algebra  $H$  and compatible equivalences*

$$\mathcal{P}_A \simeq \mathrm{Perf}(H), \quad \mathcal{P}_T \simeq \mathrm{Perf}(H \otimes_A^{\mathbf{L}} T).$$

*The equivalences respect central localization by elements of  $A$ .*

*Proof.* The sequence  $E_S = (S^{s_n})_n$  is an additive generator. Indeed, if  $a_n = O(s_n)$ , there is a single integer  $c$  such that  $a_n \leq cs_n$  for every  $n$ , after increasing  $c$  to handle the finitely many exceptional indices. Coordinate inclusions and projections make  $(S^{a_n})_n$  a summand of  $E_S^c$ . Consequently

$$\mathcal{D}_S \simeq \mathrm{Perf}(B_S), \quad B_S = \prod_n M_{s_n}(S).$$

The  $A$ -module underlying  $B_A$  is flat. One can check this by the ideal criterion: every finitely generated ideal of the Noetherian ring  $A$  is finitely presented, so its tensor product with this product of finite free modules is the corresponding product of its tensor products; each required injection is then a product of injections. Moreover,  $T$  is finitely presented over  $A$ , so tensoring with  $T$  commutes with that product. Thus

$$B_A \otimes_A^{\mathbf{L}} T \simeq B_A \otimes_A T \cong B_T. \quad (4.7)$$

The same product argument makes every morphism module between the original free sequences flat over  $A$ . Hence the explicit dg quotient of this degree-zero category by its full rank-small

subcategory computes the derived quotient [7, Section 3.1 and Theorem 3.4]. Its morphisms are sums of paths formed from tensor products of the original morphism modules and contracting homotopies in homological degree 1. They are concentrated in nonnegative degrees and degreewise  $A$ -flat, hence are  $K$ -flat. Passing to the perfect hull and idempotent completion kills exactly  $\mathcal{N}_A$ . The image of  $E_A$  remains a generator, and its endomorphism algebra  $H = \text{End}_{\mathcal{P}_A}(q_A E_A)$  is connective, giving  $\mathcal{P}_A \simeq \text{Perf}(H)$ . This assertion concerns the chosen generator, not arbitrary shifts of perfect objects.

Since  $T$  is finitely presented over  $A$ , tensoring each original morphism module with  $T$  gives the corresponding module of morphisms between free  $T$ -sequences. Tensoring the path formula therefore gives exactly the  $T$ -linear dg quotient by the rank-small free sequences. The same flatness criterion applies over the Noetherian ring  $T$ . The upstairs endomorphism algebra is consequently  $H \otimes_A T \simeq H \otimes_A^{\mathbf{L}} T$ , and its perfect hull and idempotent completion are  $\mathcal{P}_T$ . This also agrees with the preservation of exact quotients under scalar extension [18, Lemma 3.12]. All constructions are centrally  $A$ -linear and preserve central localization.  $\square$

Under these equivalences, the support condition used throughout this section is exactly the one in Theorem 3.1. Indeed, for a perfect module  $X$ , localization at a central element  $z_j$  is the telescope of multiplication by  $z_j$  in the module category. Compactness implies that  $X[1/z_j] = 0$  if and only if some power  $z_j^v$  kills its identity. A common power can be chosen for the finitely many parameters. The same argument for  $\text{Perf}(B_S)$  identifies  $\mathcal{S}_S$  with the supported objects before the rank quotient. In particular, this comparison does not require a uniformly supported lift of an object in  $\mathcal{P}_S^J$ .

Suppose now that a finite group  $G$  acts on  $T$  over  $A$ , that  $\text{Spec } T \rightarrow \text{Spec } A$  is surjective, and that  $G$  is transitive on every geometric fiber, as in Theorem 3.1. It acts on  $\mathcal{P}_T^J$  by scalar twisting. We write  $gX$  for the functor induced by  $g^*P = P \otimes_{T,g} T$ , with the convention of Section 3.

**Theorem 4.7** (Asymptotic norm comparison). *Let  $A, J, T, C, \lambda$  satisfy the setup of this section and (4.1). Suppose a finite group  $G$  acts on  $T$  over  $A$ , the map  $\text{Spec } T \rightarrow \text{Spec } A$  is surjective, and  $G$  is transitive on every geometric fiber. Then, for every  $X \in \mathcal{P}_T^J$ ,*

$$\sum_{g \in G} \chi_{\text{ord}}(gX) = \sum_{g \in G} \chi_C(gX).$$

*Neither invariance of  $\lambda$  under  $G$  nor equality of the two evaluations on individual classes is assumed.*

*Proof.* Lemma 4.6 identifies the two supported categories with the supported perfect categories of  $H$  and  $H \otimes_A^{\mathbf{L}} T$ , where  $H$  is connective. The identification over  $T$  is compatible with the  $G$ -action: scalar twisting acts on the  $T$  factor in the coefficient algebra. Theorem 3.1 therefore gives a surjection

$$K_0(\mathcal{P}_A^J) \otimes \mathbb{Q} \longrightarrow (K_0(\mathcal{P}_T^J) \otimes \mathbb{Q})^G.$$

The norm class  $\sum_g [gX]$  is invariant. The homomorphism  $\chi_{\text{ord}} - \chi_C$ , extended rationally to the Grothendieck group, vanishes on the image from  $A$  by Proposition 4.5. Applying it to a preimage of this norm class proves the assertion.  $\square$

## 5 Proof of strict positivity

We apply the two norm comparisons to different finite extensions. The first comparison replaces one of the original modules by a normalized length algebra. The second comparison is applied to bounded complexes obtained from finite approximations to a parameter quotient of that algebra. The normalized lengths need not be invariant under either Galois group.

## 5.1 Reduction and the first norm comparison

**Lemma 5.1.** *To prove Theorem 1.1, it suffices to prove the assertion for*

$$D = R/P, \quad E = R/Q, \quad h = \dim D > 0, \quad l = \dim E > 0, \quad h + l = d,$$

where  $R$  is complete with algebraically closed residue field  $k$  of characteristic  $p > 0$ ,  $P, Q$  are prime, and  $P + Q$  is primary to  $\mathfrak{m}_R$ .

*Proof.* The equicharacteristic case is Serre's positivity theorem [25]. We may therefore assume that the residue characteristic is positive. Completion, followed by the extension in Lemma 2.1, reduces to a complete regular local ring with algebraically closed residue field. For clarity, a flat local extension  $R \rightarrow R_1$  with  $\mathfrak{m}_R R_1 = \mathfrak{m}_{R_1}$  preserves the length of every finite-length module: a composition series becomes a filtration with factors  $R_1/\mathfrak{m}_{R_1}$ . Flat base change identifies the Tor modules after extension. The same argument applied to  $M/\mathfrak{m}_R^n M$  preserves the Hilbert–Samuel function, and hence the dimension of  $M$ ; it applies equally to  $N$ . It also preserves regularity, since it preserves the dimension and the minimal number of generators of the maximal ideal.

We use the vanishing theorem of Roberts and Gillet–Soulé in its general regular local form [22, 11]:

$$\chi^R(V, W) = 0 \quad \text{if} \quad \text{length}_R(V \otimes_R W) < \infty \quad \text{and} \quad \dim V + \dim W < d. \quad (5.1)$$

Take prime filtrations of  $M$  and  $N$ . All pairs of factors still have finite-length tensor product, since their supports are contained in the respective original supports. Additivity of the Euler characteristic and Equation (5.1) discard every pair except those having both original dimensions. There are top-dimensional factors in each filtration, so at least one such pair remains. Proving strict positivity for each remaining pair proves it for  $M, N$ . Their primes satisfy  $\sqrt{P + Q} = \mathfrak{m}_R$ .

Finally, a zero-dimensional prime quotient is  $k$ , and its complementary  $d$ -dimensional prime quotient is  $R$ , since a regular local ring is a domain. For that pair the characteristic equals  $\text{length}_R(k) = 1$ . Thus we may assume  $h, l > 0$ .  $\square$

Fix a pair as in Lemma 5.1. By Lemma 2.2, choose regular complete normalization bases

$$A_D \subset D, \quad A_E \subset E$$

of dimensions  $h, l$ , with residue field  $k$ , such that the extensions are finite and generically separable.

Let  $D'$  and  $E'$  be the integral closures of these bases in finite Galois closures of the respective fraction-field extensions. They are finite over their bases by excellence, and are finite over  $D, E$  as well. They are local: a finite algebra over a complete local ring is a product of complete local algebras, and a domain has only one factor [28, Tag 03QH]. Their residue fields are finite extensions of  $k$ , hence are  $k$ . Write

$$G_D = \text{Gal}(\text{Frac}(D')/\text{Frac}(A_D)), \quad G_E = \text{Gal}(\text{Frac}(E')/\text{Frac}(A_E)).$$

Normality of the bases gives  $(D')^{G_D} = A_D$  and  $(E')^{G_E} = A_E$ . For either extension write  $T^G = A$ . Let  $\Omega$  be an algebraically closed  $A$ -field and put  $S = T \otimes_A \Omega$ . For  $u = \sum_i t_i \otimes \omega_i \in S$ , the norm  $\prod_{g \in G} g(u)$  is scalar: the coefficients of  $\prod_{g \in G} (\sum_i g(t_i) X_i)$  lie in  $T^G = A$ , and we specialize  $X_i = \omega_i$ . If the finite Artinian  $\Omega$ -algebra  $S$  had more than one orbit of points, its product decomposition would give a nontrivial  $G$ -invariant idempotent  $e$ , equal to one on one orbit and zero on the others. But its norm is  $e^{|G|} = e$  and is scalar, so  $e = 0$  or  $e = 1$ , a contradiction. Thus every geometric fiber is transitive, without any assertion that invariants commute with base change. The prime

and residue-field formulation is also given in [28, Tags 0BRI and 0BRJ]. Thus the hypotheses of Theorem 3.1 apply to both extensions.

Use Theorem 2.5 to choose the algebras and normalized lengths

$$(C_D, \lambda_D) = (C_{D'}, \lambda_{D'}), \quad (C_E, \lambda_E) = (C_{E'}, \lambda_{E'}).$$

They remain fixed throughout the proof. Set  $a_D = \text{rank}_D D'$  and  $a_E = \text{rank}_E E'$ . For  $\sigma \in G_D$ , define

$$\iota_\sigma : R \longrightarrow D' \xrightarrow{\sigma} D'.$$

Give  $D'$  this  $R$ -action, and give  $C_D$  the action obtained by composing  $\iota_\sigma$  with  $D' \rightarrow C_D$ . Define

$$c_\sigma = \sum_{i=0}^d (-1)^i \lambda_D(\text{Tor}_i^R(C_D, E')), \quad (5.2)$$

where this  $R$ -action on  $C_D$  is understood.

**Lemma 5.2.** *The lengths in Equation (5.2) are finite, and*

$$|G_D| a_D a_E \chi^R(D, E) = \sum_{\sigma \in G_D} c_\sigma. \quad (5.3)$$

*Proof.* Take a finite  $R$ -free resolution of  $E'$  and extend it to  $D'$ . The resulting bounded free complex is supported at the closed point: after localizing its homology away from  $\mathfrak{m}_R$ , the supports of  $D'$  and  $E'$  are disjoint. The same holds after every twist, whose kernel on  $R$  is still  $P$ . Lemma 3.5 supplies finite normalized homology lengths and equality of the ordinary and normalized Euler evaluations after summing all conjugates.

Applying  $\sigma$  entrywise gives a semilinear isomorphism from the untwisted complex to its conjugate, so all ordinary Euler characteristics in that sum are equal to  $\chi^R(D', E')$ . Ordinary lengths over  $D'$  and over  $R$  agree here because both residue fields are  $k$ .

Choose an injection  $D^{a_D} \rightarrow D'$  with torsion cokernel. The cokernel has dimension less than  $h$ , and therefore its intersection characteristic with  $E'$  vanishes by Equation (5.1). Additivity gives

$$\chi^R(D', E') = a_D \chi^R(D, E').$$

The analogous argument for  $E \subset E'$  gives  $\chi^R(D, E') = a_E \chi^R(D, E)$ . The norm comparison now yields Equation (5.3).  $\square$

It remains to show  $c_\sigma > 0$  for each  $\sigma$ . Fix one  $\sigma$  until the final paragraph, and write  $C = C_D$  and  $\lambda = \lambda_D$  with the chosen  $R$ -action.

## 5.2 A parameter quotient and its Tor groups

Choose

$$x_1, \dots, x_h \in Q \quad (5.4)$$

which form an  $R$ -regular sequence and whose images are a system of parameters of  $D$ . Here is the simultaneous prime-avoidance argument. After  $i-1$  choices, the associated primes of  $R/(x_1, \dots, x_{i-1})$  have height  $i-1$ , since  $R$  is Cohen–Macaulay. None contains  $Q$ , whose height is  $h \geq i$ . Also avoid the primes in  $D/(x_1, \dots, x_{i-1})$  which obstruct the next parameter dimension drop. Each has positive dimension; if its inverse image contained  $Q$ , it would contain  $P+Q$  and hence would be

the maximal ideal, a contradiction. Prime avoidance inside  $Q$  therefore makes both choices possible at every step.

Put

$$B = R/(x_1, \dots, x_h), \quad I_\sigma = (\iota_\sigma(x_1), \dots, \iota_\sigma(x_h))D', \quad U = C/I_\sigma C.$$

The ideal  $I_\sigma$  is a parameter ideal of  $D'$ , since finiteness and then the automorphism  $\sigma$  preserve that property. Thus Theorem 2.5 gives

$$0 < \lambda(U) = e(I_\sigma, D') < \infty, \quad \lambda(H_q(x; C)) = 0 \quad (q > 0). \quad (5.5)$$

Here the notation  $x$  in a complex over  $C$  uses  $\iota_\sigma$ . Some integer  $a \geq 1$  satisfies

$$\mathfrak{m}_B^a(D'/I_\sigma) = 0. \quad (5.6)$$

In particular it annihilates  $U$  and every module over  $D'/I_\sigma$ .

**Lemma 5.3.** *For every finite  $B$ -module  $L$ , all the following lengths are finite, and*

$$\tau_i(L) := \lambda(\mathrm{Tor}_i^B(U, L)) = \lambda(\mathrm{Tor}_i^R(C, L)), \quad \tau_i(L) = 0 \quad (i > l). \quad (5.7)$$

*Proof.* The complexes in question have compatible  $C$ -actions. The Koszul homology  $H_q(x; C)$  is annihilated by  $x$  and by  $P$ , and is therefore bounded-primary-torsion. For  $q > 0$  its length is zero by Equation (5.5). A free  $B$ -resolution of  $L$ , with each term of finite rank, gives

$$\lambda(\mathrm{Tor}_p^B(H_q(x; C), L)) = 0 \quad (q > 0).$$

Indeed every such homology is a subquotient of a finite direct sum of a length-zero module. For  $q = 0$  these lengths are finite because  $\lambda(U) < \infty$ . The change-of-rings spectral sequence

$$\mathrm{Tor}_p^B(\mathrm{Tor}_q^R(C, B), L) \implies \mathrm{Tor}_{p+q}^R(C, L)$$

now gives the asserted equality of lengths. Only finitely many terms enter each total degree.

For the sharper bound, choose an  $R$ -free resolution  $P_L$  of  $L$  in degrees  $0, \dots, d$  and consider

$$K(x; R) \otimes_R C \otimes_R P_L.$$

Taking Koszul homology first, all positive rows have length zero, and the zero row is  $U \otimes_R P_L$ . Consequently the total homology has length zero in degrees greater than  $d$ . On the other hand,  $xL = 0$ , so in the derived category of  $R$ -modules

$$K(x; R) \otimes_R^{\mathbf{L}} L \simeq \bigoplus_{j=0}^h L[j]^{\binom{h}{j}}.$$

Tensoring this identity derivedly with  $C$  preserves the splitting as a statement about  $C$ -complexes. The total homology in degree  $i+h$  therefore contains  $\mathrm{Tor}_i^R(C, L)$  as a direct summand. If  $i > l = d-h$ , that total degree exceeds  $d$ , proving the bound. This argument also shows directly that all the displayed lengths are finite.  $\square$

### 5.3 Finite approximations preserving the actions

We need finite-length  $B$ -modules whose normalized Tor lengths approach those of  $U$ , while retaining the single annihilator  $\mathfrak{m}_B^a$ . No finite generation or finite presentation of  $U$  is assumed. The approximation will take place among modules carrying both its tower action and its  $B$ -action. In this subsection,  $\lambda$  denotes  $\lambda_\infty$  on underlying tower modules, as permitted by Theorem 2.5; the approximating modules need not carry a  $C$ -action.

Let  $A = A_D$ , let  $A_j$  be its flat regular tower from Lemma 2.3, and let  $\Lambda = \bigcup_j A_j$ . Thus

$$s(j) := \text{rank}_A A_j = p^{jh}.$$

Set

$$S_j = A_j \otimes_A (D'/I_\sigma), \quad S = \Lambda \otimes_A (D'/I_\sigma). \quad (5.8)$$

The commuting  $\Lambda$ - and  $D'/I_\sigma$ -actions on  $U$  combine into an  $S$ -action. Working with  $S$ -modules retains both of these actions. Descent to  $S_j$  retains the existing  $B$ -action by restriction along  $B \rightarrow D'/I_\sigma$  induced by  $\iota_\sigma$ ; no new  $B$ -action is chosen at a finite stage.

**Lemma 5.4.** *There are finite-length  $B$ -modules  $M_n$  and positive integers  $s_n \rightarrow \infty$ , with  $\mathfrak{m}_B^a M_n = 0$ , such that*

$$\frac{\text{length}_B(M_n)}{s_n} \rightarrow \lambda(U), \quad (5.9)$$

$$\frac{\text{length}_B \text{Tor}_i^B(M_n, L)}{s_n} \rightarrow \tau_i(L) \quad (L = k, E', i \geq 0). \quad (5.10)$$

Writing  $\beta_0(M_n) = \dim_k(M_n/\mathfrak{m}_B M_n)$ , we also have

$$\frac{\beta_0(M_n)}{s_n} \rightarrow \nu_\sigma, \quad \nu_\sigma \geq \frac{\lambda(U)}{\text{length}_B(B/\mathfrak{m}_B^a)} > 0. \quad (5.11)$$

*Proof.* The algebra  $S$  is finitely presented as a  $\Lambda$ -module and has finite normalized length. Every  $S_j$  is finite over the twisted image of  $B$ :  $D'$  is finite over the twisted image of  $R/P$ , and  $A_j$  is finite free over  $A$ . Equation (5.6) annihilates  $S_j$  and  $S$ . Moreover,  $S_j$  is Artinian local with residue field  $k$ . To see locality, the tower parameter roots are nilpotent over the Artinian quotient  $D'/I_\sigma$ , and reduction leaves the common residue field  $k$ . Thus finite  $S_j$ -modules have finite lengths over both  $B$  and  $A_j$ , and these lengths agree: a composition series over  $S_j$  has factors  $k$ , of length one for either ring. This argument is unchanged if a residue-field identification is composed with an automorphism.

Given  $\epsilon > 0$ , choose a finitely generated  $S$ -submodule  $V \subset U$  with  $\lambda(U/V) < \epsilon$ . This follows from the supremum property of normalized length: the  $S$ -span of a finitely generated  $\Lambda$ -submodule is still finitely generated over  $\Lambda$ . To obtain a finite presentation as well, choose a surjection  $S^r \rightarrow V$  with kernel  $K$ . Its kernel has finite length, bounded by  $r\lambda(S)$ . Choose a finitely generated  $S$ -submodule  $K_0 \subset K$  such that  $\lambda(K/K_0) < \epsilon$ . Then

$$P_\epsilon = S^r/K_0 \rightarrow U$$

has kernel and cokernel of length less than  $\epsilon$ . It is finitely presented over  $S$  and over  $\Lambda$ , and is still annihilated by  $\mathfrak{m}_B^a$ .

Fix  $B$ -resolutions of  $k$  and  $E'$  with finite free terms, and write  $r_i(L)$  for the rank in degree  $i$ , with  $r_{-1}(L) = 0$ . If  $K_\epsilon$  and  $W_\epsilon$  are the kernel and cokernel of  $P_\epsilon \rightarrow U$ , the two long exact homology sequences give

$$\begin{aligned} & \left| \lambda \text{Tor}_i^B(P_\epsilon, L) - \lambda \text{Tor}_i^B(U, L) \right| \\ & \leq (r_i(L) + r_{i-1}(L))\lambda(K_\epsilon) + (r_{i+1}(L) + r_i(L))\lambda(W_\epsilon). \end{aligned} \quad (5.12)$$

For example, each homology of the complex obtained by tensoring the resolution with  $K_\epsilon$  has length at most the rank of the corresponding term times  $\lambda(K_\epsilon)$ ; the same holds for  $W_\epsilon$ . All these quantities are finite. Also  $|\lambda(P_\epsilon) - \lambda(U)| < 2\epsilon$ .

A finite  $S$ -presentation of  $P_\epsilon$  descends to an  $S_j$ -module  $P_j$  for some  $j$ , with  $P_\epsilon = \Lambda \otimes_{A_j} P_j$ . Flatness of  $\Lambda/A_j$  gives, for every  $i$ ,

$$\mathrm{Tor}_i^B(P_\epsilon, L) = \Lambda \otimes_{A_j} \mathrm{Tor}_i^B(P_j, L).$$

The finite-stage homology on the right is finite over  $S_j$ . The normalization formula in Theorem 2.5, and the length comparison above, therefore give exactly

$$\lambda \mathrm{Tor}_i^B(P_\epsilon, L) = \frac{\mathrm{length}_{A_j} \mathrm{Tor}_i^B(P_j, L)}{s(j)} = \frac{\mathrm{length}_B \mathrm{Tor}_i^B(P_j, L)}{s(j)}.$$

The same formula holds for  $P_\epsilon$  itself.

At step  $n$ , choose  $\epsilon$  so that all errors in Equation (5.12), for  $i \leq n$  and  $L = k, E'$ , and the length error are less than  $1/n$ . Descend this presentation and enlarge its stage if necessary so that  $j \geq n$ . Further flat tower extension leaves every normalized value unchanged. Take  $M_n = P_j$  and  $s_n = s(j)$ . This proves Equations (5.9) and (5.10), including the common annihilator and  $s_n \rightarrow \infty$ .

The case  $i = 0, L = k$  gives the limit in Equation (5.11). A minimal generating set and the annihilator give a surjection  $(B/\mathfrak{m}_B^a)^{\beta_0(M_n)} \rightarrow M_n$ . Consequently

$$\mathrm{length}_B(M_n) \leq \beta_0(M_n) \mathrm{length}_B(B/\mathfrak{m}_B^a).$$

Divide by  $s_n$  and take limits to obtain the stated positive lower bound.  $\square$

## 5.4 Bounded complexes and their ordinary evaluation

The approximations provide two kinds of control. In minimal resolutions, the Tor bound  $\tau_i(k) = 0$  for  $i > l$  makes the corresponding ranks negligible. The common annihilator then makes the truncated sequences supported after quotienting by negligible ranks. That same annihilator has already given the positive normalized generator bound (5.11), which will make the second evaluation strictly positive.

Choose minimal free  $B$ -resolutions  $F_n^\infty$  of  $M_n$ , indexed homologically in nonnegative degrees. Let  $F_n$  be their brutal truncations to degrees  $0, \dots, l$ . Minimality and Equation (5.10) give, in each fixed degree,

$$\mathrm{rank}_B F_{n,i}^\infty = O(s_n), \quad \mathrm{rank}_B F_{n,i}^\infty = o(s_n) \quad (i > l), \quad (5.13)$$

because these ranks equal  $\mathrm{length}_B \mathrm{Tor}_i^B(M_n, k)$ .

Let  $z_1, \dots, z_l$  be regular parameters of  $A_E$ . For  $\gamma \in G_E$ , extend  $F_n$  through

$$B \longrightarrow E \longrightarrow E' \xrightarrow{\gamma} E',$$

and denote the resulting complex by  $F'_{n,\gamma}$ . Use the normalization  $s_n$  in the asymptotic category over  $E'$ . These numbers come from the  $D$ -side approximation. Section 4 allows any positive integer sequence  $s_n$ , so no identification or compatibility between the parameter-root towers of  $A_D$  and  $A_E$  is needed.

**Lemma 5.5.** *The sequence  $(F'_{n,\gamma})_n$  is supported at the closed point in that category. Its ordinary Euler evaluation is  $c_\sigma$  for every  $\gamma \in G_E$ .*

*Proof.* Every element  $b \in \mathfrak{m}_B^a$  acts nullhomotopically on  $F_n^\infty$ , because it annihilates  $M_n$ . Restrict a chosen homotopy to the brutal truncation. Its only remaining error is in degree  $l$ , where it is

$$F_{n,l} \xrightarrow{h_l} F_{n,l+1}^\infty \xrightarrow{d_{l+1}} F_{n,l}. \quad (5.14)$$

This is a factorization by chain maps through  $F_{n,l+1}^\infty[l]$ : the map induced by  $d_{l+1}$  is a chain map because  $d_l d_{l+1} = 0$ , and the map induced by  $h_l$  is a chain map because the truncation has no degree  $l+1$  term. The middle module has rank  $o(s_n)$  by Equation (5.13), and is therefore killed in the asymptotic quotient. These identities persist after extension to  $E'$ , without a flatness assumption over  $B$ .

The ideal  $\mathfrak{m}_B E'$  for each indicated structural map is primary to  $\mathfrak{m}_{E'}$ . Choose  $v$  so large that every  $z_j^v$  belongs to the extension of  $\mathfrak{m}_B^a$  for all  $\gamma$ . This is one fixed choice, since the group is finite. Expressing  $z_j^v$  as a finite  $E'$ -linear combination of elements of  $\mathfrak{m}_B^a$  proves that it acts by zero on the sequence in the quotient. The same combinations give genuine nullhomotopies on the full extended resolutions. This proves support.

Apply the ordinary version of Equation (4.2). In its right-hand side, one may replace the truncated extended resolutions by the full ones in each fixed homological degree. Indeed, the quotient of the full resolution by its brutal truncation has only terms of degrees at least  $l+1$ , all of rank  $o(s_n)$  in each fixed degree. After tensoring that tail with  $K(z^v; E')$ , taking Koszul homology first bounds each fixed-degree homology by finite sums of

$$\text{rank}_B F_{n,i}^\infty \text{ length}_{E'} H_q(z^v; E') = o(s_n).$$

The Koszul homology lengths are fixed and finite, and only finitely many terms contribute to each total degree. The long exact homology sequence thus proves the replacement assertion.

On each full extended resolution the powers  $z_j^v$  act nullhomotopically, as established above. Koszul tensoring therefore gives the direct sum of shifts with multiplicities  $\binom{l}{j}$ . Its underlying homology has ordinary lengths  $\text{length}_B \text{Tor}_i^B(M_n, E')$ , unchanged by conjugating the  $E'$ -action: applying  $\gamma$  entrywise is a semilinear isomorphism. Equations (5.10) and (5.7), followed by coefficient comparison in Equation (4.2), now identify the ordinary homology-length coefficients with

$$\tau_i(E') = \lambda \text{Tor}_i^R(C, E').$$

Their alternating sum is exactly  $c_\sigma$ . □

## 5.5 A positive normalized evaluation

The ordinary evaluation has recovered  $c_\sigma$  on every conjugate. We now evaluate the same supported object using  $C_E$ . The parameter Koszul identities will leave only its nonnegative degree-zero coefficient, and minimality of the resolutions will bound that coefficient below using (5.11).

Corollary 2.6 gives

$$0 < \eta_E := \lambda_E(C_E / \mathfrak{m}_{E'} C_E) < \infty. \quad (5.15)$$

We record a quantitative bound for use below. Set  $q_E = (z_1, \dots, z_l)E'$ ,  $r_E = \text{rank}_{A_E} E'$ , and  $t_E = \text{length}_{E'}(E'/q_E)$ . The length input gives  $\lambda_E(C_E/q_E C_E) = r_E > 0$ . A composition series of  $E'/q_E$ , tensored right-exactly with  $C_E$ , shows

$$r_E \leq t_E \eta_E. \quad (5.16)$$

At each step the image of the tensor of the submodule has length at most that tensor's length; injectivity after tensoring is not required. Finiteness of  $\eta_E$  follows from the surjection  $C_E/q_E C_E \rightarrow C_E/\mathfrak{m}_{E'} C_E$ .

**Lemma 5.6.** *The  $\lambda_E$  Euler evaluation of  $(F'_{n,\gamma})_n$  is a number  $b_{\sigma,\gamma}$  satisfying*

$$b_{\sigma,\gamma} \geq \nu_\sigma \eta_E \geq \frac{\lambda_D(U)}{\text{length}_B(B/\mathfrak{m}_B^a)} \frac{r_E}{t_E} > 0. \quad (5.17)$$

*Proof.* Let  $b_i$  be the nonnegative homology-length coefficients of this evaluation in Equation (4.2). Consider the finite double complex

$$K(z^v; E') \otimes_{E'} F'_{n,\gamma} \otimes_{E'} C_E.$$

Taking Koszul homology first, Theorem 2.5 makes all positive Koszul rows have length zero. The remaining row has complex degrees  $0, \dots, l$ . Thus the normalized homology lengths of this double complex vanish in total degrees greater than  $l$ , as well as in negative degrees. Equation (4.2) says that their polynomial is  $(1+t)^l \sum_i b_i t^i$ . All  $b_i$  are nonnegative. Boundedness and comparison of the least degree rule out negative-degree coefficients. Any  $b_i > 0$  with  $i > 0$  would then give a positive coefficient in degree  $i+l > l$ , a contradiction. Consequently only  $b_0$  can be nonzero, and the Euler evaluation equals  $b_0$ .

In degree zero, the displayed double complex has homology

$$C_E^{\beta_0(M_n)} / \left( \text{im}(d_1) + (z_1^v, \dots, z_l^v) C_E^{\beta_0(M_n)} \right).$$

Since  $l > 0$ ,  $d_1$  is the degree-one presentation differential of  $M_n$ . Minimality over  $B$  puts all its entries in  $\mathfrak{m}_B$ , whose image under any twist is contained in  $\mathfrak{m}_{E'}$ . The base parameter powers are also in  $\mathfrak{m}_{E'}$ . There is therefore a surjection from this degree-zero homology onto

$$(C_E/\mathfrak{m}_{E'} C_E)^{\beta_0(M_n)}.$$

Its normalized length is at least  $\beta_0(M_n) \eta_E / s_n$ . The degree-zero coefficient in Equation (4.2) is  $b_0$ , so taking the ultralimit and using the ordinary limit in Equation (5.11) gives  $b_0 \geq \nu_\sigma \eta_E$ . Equations (5.11) and (5.16) give the explicit remaining inequalities in Equation (5.17). Define  $b_{\sigma,\gamma} = b_0$ .  $\square$

*Proof of Theorem 1.1.* By Lemma 5.1, use the domain pair fixed above. For each  $\sigma \in G_D$ , carry out the construction with its specified  $R$ -action. Apply Theorem 4.7 to the supported sequence of Lemma 5.5, over  $A_E \subset E'$ . It equates the sums of the ordinary and normalized evaluations of all  $G_E$ -conjugates. The former evaluation is  $c_\sigma$  for every conjugate, whereas the latter is  $b_{\sigma,\gamma}$ . Hence

$$|G_E| c_\sigma = \sum_{\gamma \in G_E} b_{\sigma,\gamma} > 0 \quad (5.18)$$

by Lemma 5.6. Finally, Lemma 5.2 gives

$$|G_D| a_D a_E \chi^R(D, E) = \frac{1}{|G_E|} \sum_{\sigma \in G_D} \sum_{\gamma \in G_E} b_{\sigma,\gamma} > 0.$$

Both ranks and both group orders are positive. Thus  $\chi^R(D, E) > 0$ , and the reduction proves the theorem. Neither norm comparison requires a Galois action on  $C_D$  or  $C_E$ , nor invariance of either normalized length under a twist.  $\square$

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